LINEAR ALGEBRAIC GROUPS

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Abstract. These are notes for a graduate course on linear algebraic groups taught at Zhejiang University. They were written by Zhu Ruizhi during class for which I thank him sincerely. Any comments are highly appreciated; please send them directly to mihatsch@zju.edu.cn.

Contents

1. Introduction (9/16/2026) 2
1.1. What is the theory about? 2
1.2. Affine varieties 4
1.3. Linear algebraic groups 8
2. Topological properties (9/23/2026) 10
2.1. Examples 10
2.2. Topological properties 14
2.3. Connectedness for LAGs 18
3. Embedding into GL_n and images of morphisms (9/30/2026) 20
3.1. Embedding into GL_n. 20
3.2. Varieties 23
3.3. Constructible subsets 24
3.4. Images are again LAGs 27
4. Actions and orbits (10/10/2026) 27
4.1. Orbits 28
4.2. Natural examples of group actions 31
5. Two case studies 35
5.1. Case study 1: The nilpotent cone in $\mathrm{M}_n(k)$ 36
5.2. Case study 2: $\mathrm{Hom}(\mathrm{GL}_n, \mathbb{G}_a)$ 40
6. Projective varieties and quotients (10/14/2026) 40
6.1. Projective varieties 40
6.2. Products 42
6.3. Grassmannians 43
6.4. Constructing quotients as varieties 45
6.5. The quotient property 46
7. The Lie algebra (10/21/2026) 48
7.1. Motivation 48
7.2. Vector fields and tangent spaces 49
7.3. Lie algebras 51
References 53

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1. Introduction (9/16/2026)

Course Information

- Prerequisites: Commutative Algebra or Introduction to Algebraic Geometry
- One lecture per week, on Wednesdays 10:00 a.m. to 12:30 p.m.
- Homework graded for feedback, no effect on final grade
- Two exams:
  - 1st on November 4th, 40%.
  - 2nd on December 30th, 60%.
- Main reference: Humphrey's Linear Algebraic Groups [1]

The goal of this first lecture are to explain what the theory of linear algebraic groups is about, to introduce affine varieties, and to give the definition of a linear algebraic group.

1.1. **What is the theory about?** Mathematicians and physicist are interested in groups such as GL_n, SL_n, SO_n, or Sp_{2n}. These arise in many different contexts and with different flavors:

(1) Finite groups of Lie type $\mathrm{GL}_n(\mathbb{F}_q), \mathrm{Sp}_{2n}(\mathbb{F}_q)$,... come from a more set-theoretic, combinatorial perspective. They make up part of classification of the finite simple groups.
(2) Real or complex Lie groups $\mathrm{GL}_n(\mathbb{R}), \mathrm{SO}_n(\mathbb{C}), \mathrm{SL}_n(\mathbb{R}), \mathrm{PGL}_n(\mathbb{C})$,... arise from a more analytic perspective which is relevant in differential geometry, representation theory, etc.

Our perspective. We want to study these groups as algebraic objects. An advantage of this approach is that it makes sense over various fields. For example, we can study $\mathrm{SO}(V)/\mathbb{Q}, \mathrm{GL}_n/\mathbb{Q}_p, \mathrm{Sp}_{2n}/\mathbb{C}$ which is clearly relevant for areas such as number theory, commutative algebra, algebraic geometry, and so on.

Example 1.1. We consider $\mathrm{SL}_2/\mathbb{C}$ as algebraic or complex Lie group, and are interested in its representation theory.

By definition, a representation of SL_2 is a finite-dimensional $\mathbb{C}$-vector space V together with a homomorphism $\mathrm{SL}_2 \to \mathrm{GL}(V)$. Such representations can be completely classified:

Theorem 1.2. *Let V be the finite-dimensional representation of $\mathrm{SL}_2/\mathbb{C}$.*

(a) V is completely reducible*, meaning it is a direct sum of irreducible representations,*

$$V \cong \bigoplus_{i=1}^{r} V_i.$$

(b) Each irreducible summand V_i is isomorphic to the n-th symmetric power $\mathrm{Sym}^n(\mathbb{C}^2)$ of the standard representation, for some n.

Explanation. What is $\mathrm{Sym}^n(\mathbb{C}^2)$?

(1) $\mathrm{SL}_2 \curvearrowright \mathbb{C}^2$ is the *standard representation*

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}. \tag{1.1}$$

(2) $\mathrm{Sym}^n(\mathbb{C}^2)$ is derived from $\mathbb{C}^2$ as the space of homogeneous degree n polynomials in x and y:

$$\mathrm{Sym}^n(\mathbb{C}^2) = \{a_n x^n + a_{n-1} x^{n-1} y + \cdots + a_0 y^n \mid a_n, \ldots, a_0 \in \mathbb{C}\}.$$

SL_2 acts by the coordinate substitution given by (1.1):

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot p(x,y) = p(ax+by, cx+dy).$$

For example,

$$\begin{pmatrix} 1 & 1 \\ & 1 \end{pmatrix} \cdot x^2 = (x+y)^2 = x^2 + 2xy + y^2.$$

Theorem 1.2 is a beautiful example of a complete classification result. But such behavior is not automatic for algebraic groups, as the next example shows.

Example 1.3. Consider the *additive group* $\mathbb{G}_a = (\mathbb{C}, +)$. It has the two-dimensional representation

$$\mathbb{G}_a \longrightarrow \mathrm{GL}_2, \qquad t \longmapsto \begin{pmatrix} 1 & t \\ & 1 \end{pmatrix}.$$

- This representation is not irreducible: $\mathbb{C} \cdot \left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right) \subseteq \mathbb{C}^2$ is a $\mathbb{G}_a$-stable subspace.
- But $\mathbb{C} \cdot \left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$ has no invariant complement:

$$\left\langle \begin{pmatrix} a \\ b \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ & 1 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a+b \\ b \end{pmatrix} \right\rangle = \mathbb{C}^2 \qquad \text{if } b \neq 0.$$

Question. Which groups behave like SL_2, which like $\mathbb{G}_a$?

General theory answers this question:

- *reductive groups* behave like SL_2 (completely reducible representation theory)
- *unipotent groups* behave like $\mathbb{G}_a$ (every rep has a fixed vector like $\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right) \in \mathbb{C}^2$ before)

Example 1.4. The groups $\mathrm{GL}_n, \mathrm{SL}_n, \mathrm{PGL}_n, \mathrm{SO}_n, \mathrm{Sp}_{2n}$ are all reductive.

Upper triangular matrices form a unipotent group, e.g.

$$\left\{ \begin{pmatrix} 1 & x & y \\ & 1 & z \\ & & 1 \end{pmatrix} \right\} \subseteq \mathrm{GL}_3.$$

Question. OK, but I've already seen all these groups before. Can you teach me some new ones?

Answer. General theory says that these examples are the building blocks for *all* LAGs!

Theorem 1.5. *Let $G/\mathbb{C}$ be a connected linear algebraic group.*

(a) There exists a unique maximal, normal, unipotent subgroup $G_u \subseteq G$ called the unipotent radical *of G. The quotient $G_{\mathrm{red}} = G/G_u$ is reductive.*

(b) The unipotent part G_u is built as a successive extension of copies of $\mathbb{G}_a$. The reductive quotient G_{red} is built out of $\mathrm{GL}_n, \mathrm{SL}_n, \mathrm{SO}_n$ and Sp_{2n}.

Example 1.6. Consider the block upper triangular subgroup

$$G := \left\{ \begin{pmatrix} x_{11} & x_{12} & y \\ x_{21} & x_{22} & z \\ 0 & 0 & x_{33} \end{pmatrix} \right\} \subseteq \mathrm{GL}_3.$$

Then

$$G_u = \left\{ \begin{pmatrix} 1 & 0 & y \\ & 1 & z \\ & & 1 \end{pmatrix} \right\} \cong \mathbb{G}_a^2, \qquad G_{\mathrm{red}} \cong \mathrm{GL}_2 \times \mathrm{GL}_1.$$

Example 1.7. The *projective linear group* PGL_n is reductive and obtained as quotient

$$\mathrm{SL}_n \xrightarrow{n:1} \mathrm{PGL}_n.$$

The *spinor group* Spin_n is reductive and obtained as a double cover

$$\mathrm{Spin}_n \xrightarrow{2:1} \mathrm{SO}_n, \qquad n \geq 3.$$

The general theory also more precisely classifies all reductive groups and their representations up to isomorphism, which we will see towards the end of the course.

To end this introduction, let us consider a different type of statement.

Question. How to work with group elements "like a professional"?

Example 1.8. Any $g \in \mathrm{GL}_n(\mathbb{C})$ can be written uniquely as a product $g = g_s \cdot g_u$ such that

- g_s is diagonalizable (also called *semi-simple*)
- g_u is unipotent (meaning $g_u - 1 \in \mathrm{M}_n(\mathbb{C})$ is nilpotent)
- $g_s \cdot g_u = g_u \cdot g_s$

Indeed, we know that g has a unique *Jordan normal form* up to permutation of blocks, and the decomposition $g = g_s \cdot g_u$ is immediate to see in JNF. For example,

$$g = \begin{pmatrix} 2 & 5 & \\ & 2 & \\ & & -1 \end{pmatrix} = \underbrace{\begin{pmatrix} 2 & & \\ & 2 & \\ & & -1 \end{pmatrix}}_{g_s} \cdot \underbrace{\begin{pmatrix} 1 & 5/2 & \\ & 1 & \\ & & -1 \end{pmatrix}}_{g_u}.$$

Here, note that g_s and g_u indeed commute as required.

Theorem 1.9. *Let G be a connected LAG over $\mathbb{C}$. Any $g \in G$ has a unique decomposition $g = g_s g_u$ into a semi-simple and unipotent part, characterized by*

$$\forall\ \rho : G \longrightarrow \mathrm{GL}_n, \qquad \rho(g) = \rho(g_s) \cdot \rho(g_u)$$

is the decomposition from Example 1.8 $\rho(g)$.

Conclusion. Summing up this lecture, we have seen that the theory of LAGs

- Classifies and describes algebraic groups such as $\mathrm{GL}_n, \mathrm{SL}_n, \mathrm{Spin}_n$,
- Classifies and describes their representations, and
- Provides structural tools for working with such groups or their group elements.

1.2. **Affine varieties.** Reference: [1, §1.1-1.5]

In order to define linear algebraic groups, we first need to explain how to capture the idea of an *algebraic* object. This is done by the notion of an *affine algebraic variety*.

Let k be an algebraically closed field.

Slogan. An affine variety over k is the vanishing set in some k^n defined by a family of polynomial equations.

We now make this definition precise. Throughout, let

$$A \cong k[T_1, \ldots, T_n]/(f_1, \ldots, f_m)$$

be a reduced finite type k-algebra. We want to define the affine variety $\mathrm{Sp}(A)$ attached to A, which will be done in three steps.

1st Step: The underlying set. Define the *Max-Spectrum* of A as

$$\mathrm{Sp}(A) := \{\text{maximal ideals of } A\}.$$

Because k is assumed algebraically closed, for every $\mathfrak{m} \in \mathrm{Sp}(A)$, the residue field of $\mathfrak{m}$ is naturally identified with k:

$$\begin{array}{ccc} k & \longrightarrow & A \\ & \searrow^{\cong} & \downarrow \text{quotient} \\ & & A/\mathfrak{m} \end{array} \qquad \text{(also use } A \text{ is a } k\text{-algebra.)}$$

So we see:

$$\begin{aligned} \mathrm{Sp}(A) &= \{k\text{-alg homom } A \longrightarrow k\} \\ &= \{(x_1, \ldots, x_n) \in k^n \mid f_1(x_1, \ldots, x_n) = \cdots = f_m(x_1, \ldots, x_n) = 0\}. \end{aligned}$$

choosing presentation $A = k[T_1, \ldots, T_n]/(f_1, \ldots, f_m)$

Example 1.10.

$$\mathrm{Sp}(k[T_1, \ldots, T_n]) = k^n$$

$$\mathrm{Sp}(k[X, Y]/(XY)) = \quad \subseteq k^2$$

$$\mathrm{Sp}(k[X, Y]/(Y^2 - X^2(X+1))) = \quad \subseteq k^2$$

2nd Step: Upgrade to a toplogical space. Every $f \in A$ defines a function $\mathrm{Sp}(A) \longrightarrow k$

$$\begin{array}{ccc} k & \longrightarrow & A \\ & = & \downarrow \mathfrak{m} \in \mathrm{Sp}(A) \\ & & k \end{array} \qquad \begin{array}{c} f \\ \downarrow \\ f \bmod \mathfrak{m}. \end{array}$$

We usually write $x = \mathfrak{m} \in \mathrm{Sp}(A)$ and $f(x) := f \bmod \mathfrak{m} \in k$ where $k = A/\mathfrak{m}$, because this is much more intuitive notation.

Polynomial functions should definitely be continuous, but k has no topology. So the best one can do in algebraic geometry is to *define* open and closed subsets by

$$\begin{aligned} V(f) &= \{x \in \mathrm{Sp}(A) \mid f(x) = 0\} \quad \text{closed "vanishing set"} \\ D(f) &= \{x \in \mathrm{Sp}(A) \mid f(x) \neq 0\} \quad \text{open "doesn't vanish set".} \end{aligned} \tag{1.2}$$

Definition 1.11. Endow $\mathrm{Sp}(A)$ with topology generated by $\{D(f)\}_{f \in A}$.

What does this mean concretely? We already have the following opens from (1.2):

$$\mathrm{Sp}(A) = D(1), \qquad \varnothing = D(0)$$

$$D(f_1) \cap \cdots \cap D(f_r) = D(f_1 \cdots f_r)$$

So the potential new open subsets are the arbitrary unions $\bigcup_{i \in I} D(f_i)$ of sets of the form $D(f)$. Note that an open of the form $D(f)$ is called *principal open subset.*

But A is noetherian: Ther exist finitely many $g_1, \ldots, g_r$ generating the ideal $(f_i)_{i \in I}$. Then

$$\bigcap_{i \in I} V(f_i) = \bigcap_{j=1}^{r} V(g_j) \quad \text{or equivalently,} \quad \bigcup_{i \in I} D(f_i) = \bigcup_{j=1}^{r} D(g_j).$$

Conclusion. The open subsets of $\mathrm{Sp}(A)$ are precisely the finite unions $\bigcup_{i=1}^{r} D(f_i)$.

Dually, the closed subsets are precisely the finite intersections

$$V(f_1, \dots, f_r) := \bigcap_{i=1}^{r} V(f_i).$$

Next, we want to connect this topology closer to the ring A. We in fact have back-and-forth definitions:

$$A \supseteq S \text{ subset } \longmapsto V(S) := \bigcap_{s \in S} V(s),$$
$$I(Z) := \{f \in A \mid f(z) = 0 \ \forall z \in Z\} \longleftarrow Z \subseteq \mathrm{Sp}(A) \text{ closed}.$$

You should convince yourself of the following easy properties:

- $V(S)$ is closed
- $I(Z)$ an ideal
- $V(S) = V((S))$, $(S) \subseteq A$ ideal generated by S
- If $f^r \in I(Z)$, then $f \in I(Z)$. That is, $I(Z)$ is a so-called *radical* ideal. For example, $V(f^r) = V(f)$.

Theorem 1.12 (Hilbert's Nullstellensatz)**.** *The above maps define a bijection*

$$\{\textit{radical } I \subseteq A\} \overset{1:1}{\longleftrightarrow} \{\textit{closed } Z \subseteq \mathrm{Sp}(A)\}$$
$$I \longmapsto V(I),$$
$$I(Z) \longleftarrow Z.$$

The significance of this theorem is that it allows to translate any statement about unions, intersections, non-emptyness etc. of closed or open subsets of $\mathrm{Sp}(A)$ into a statement about ideals in A, and vice versa. That is, it provides a dictionary between geometry and algebra.

3rd Step: Equip $\mathrm{Sp}(A)$ **with its polynomial functions.** Set $X = \mathrm{Sp}(A)$ in the following. For every open $U \subseteq X$, we want to say what the polynomial functions on U are. This is similar to the definition of the smooth functions on an open subset of a manifold, for example.

For traditional reasons, what we intuitively just called "polynomial" functions, is usually called *regular functions*.

Definition 1.13. For $U \subseteq \mathrm{Sp}(A)$ open, we denote by $\mathscr{O}_X(U)$ the ring of all regular functions $f : U \to k$.

(1) If $U = D(h)$ is principal open, then by definition

$$\mathscr{O}_X(D(h)) := A[h^{-1}].$$

Note that this makes sense since $h : D(h) \to k^\times$ is an invertible function.

(2) For general U, we say that $f : U \to k$ is regular if $\forall D(h) \subseteq U$, the restriction $f|_{D(h)} \in \mathscr{O}_X(D(h))$.

Definition 1.14. The pair $(\mathrm{Sp}(A), \mathscr{O}_{\mathrm{Sp}(A)})$ is the *affine variety* defined by A. Here, $\mathrm{Sp}(A)$ denotes the topological space, and $\mathscr{O}_{\mathrm{Sp}(A)}$ denotes the regular functions on $\mathrm{Sp}(A)$. But usually one only writes $\mathrm{Sp}(A)$ for brevity.

Example 1.15. The *n-dimensional affine space* over k is

$$\mathbb{A}^n := \mathrm{Sp}\, k[T_1, \dots, T_n].$$

(1) Underlying set $= k^n$,
(2) Opens = complements of vanishing sets $V(f_1, \dots, f_m)$,

(3) Regular functions =

$$\mathscr{O}_{\mathbb{A}^n}(U) = \left\{ \frac{p}{q} \in k(T_1, \ldots, T_n) \;\middle|\; q(x) \neq 0 \ \forall x \in U \right\}.$$

Example 1.16. Here is an interesting example of regular functions on a non-principal open: Consider $\mathbb{A}^2 = \operatorname{Sp} k[X, Y]$ with open subset $U = k^2 \setminus \{(0,0)\}$. This is the union of $D(X)$ and $D(Y)$. Then

$$\mathscr{O}_{\mathbb{A}^2}(U) = k[X^{\pm 1}, Y] \cap k[X, Y^{\pm 1}] = k[X, Y]$$

where the intersection happens in the **quotient field** $k(X, Y) = \operatorname{Quot}(k[X, Y])$ (also called the **fraction field**).

The above procedure is in fact general: If A is an integral domain, then

$$\mathscr{O}_{\operatorname{Sp}(A)}(U) = \bigcap_{D(f) \subseteq U} A[f^{-1}], \quad \text{intersect in } \operatorname{Quot}(A).$$

It suffices to take the intersection over an open covering $U = D(f_1) \cup \ldots \cup D(f_m)$ in fact.

4th Step: Morphisms. We had three steps to define the object $\operatorname{Sp}(A)$, but we also need to define morphisms between affine varieties. Let $X = \operatorname{Sp}(A)$ and $Y = \operatorname{Sp}(B)$ in the following.

Definition 1.17. A morphism from Y to X is a map $\varphi : Y \to X$ that preserves regular functions. That is, if $V \subseteq Y$ and $U \subseteq X$ are open with $\varphi(V) \subseteq U$, then for every $f \in \mathscr{O}_X(U)$, the composition

$$f \circ \varphi : V \longrightarrow k$$

lies in $\mathscr{O}_Y(V)$.

• In particular, if $\varphi : Y \to X$ is a morphism, we may apply this definition to $V = Y$ and $X = U$ and obtain a ring map

$$\begin{aligned} \varphi^* : A = \mathscr{O}_X(X) &\longrightarrow \mathscr{O}_Y(Y) = B \\ f &\longmapsto f \circ \varphi. \end{aligned}$$

This is the *pullback map* on the coordinate rings of X and Y.

• Conversely, any k-algebra homomorphism $\varphi^* : A \longrightarrow B$ k induces a morphism of affine varieties

$$\begin{aligned} \varphi = \operatorname{Sp}(\varphi^*) : Y &\longrightarrow X \\ B \supseteq \mathfrak{m}_B &\longmapsto (\varphi^*)^{-1}(\mathfrak{m}_B) \subseteq A. \end{aligned}$$

Theorem 1.18 (Hilbert's Nullstellensatz)**.** *The above functors define an equivalence of categories*

$$\begin{aligned} \left\{ \begin{array}{c} \textit{reduced finite type} \\ \textit{k-algebras} \end{array} \right\}^{\mathrm{op}} &\xrightarrow{\cong} \left\{ \begin{array}{c} \textit{affine varieties} \\ \textit{over } k \end{array} \right\} \\ A &\longmapsto \operatorname{Sp}(A), \\ \mathscr{O}_X(X) &\longleftarrow\!\shortmid\ (X, \mathscr{O}_X). \end{aligned}$$

The general principle for passing between maps of varieties and algebra homomorphisms is the following. A morphism

$$k^n \xrightarrow{\varphi} k^m$$

given by an m-tuple of polynomials in n variables

$$(x_1, ..., x_n) \longmapsto (\varphi_1(x_1, ..., x_n), ..., \varphi_m(x_1, ..., x_n))$$

corresponds to the ring map

$$\begin{aligned} k[X_1, ..., X_n] &\xrightarrow{\widetilde{\varphi}} k[T_1, ..., T_m] \\ \varphi_i(X_1, ..., X_m) &\longleftarrow\!\shortmid T_i. \end{aligned}$$

1.3. **Linear algebraic groups.** Reference: [1, §7.1, §7.2]

Throughout, k is an algebraically closed field of characteristic 0. We often abbreviate "linear algebraic group" by LAG.

Definition 1.19. An LAG over k is an affine variety G over k together with a group structure

$$m : G \times G \longrightarrow G$$

which is also a morphism of varieties.

Explanation. Let $X = \mathrm{Sp}(A)$, $Y = \mathrm{Sp}(B)$ be affine varieties over k. Then $X \times Y$ is the affine k-variety $\mathrm{Sp}(A \otimes_k B)$.

Why does this make sense?

- Mathematically:

$$\begin{aligned} A = k[T_1, \ldots, T_n]/(f_1, \ldots, f_m) &\longleftrightarrow X = V(f_1, \ldots, f_m) \subseteq k^n, \\ B = k[S_1, \ldots, S_r]/(g_1, \ldots, g_s) &\longleftrightarrow Y = V(g_1, \ldots, g_s) \subseteq k^r. \end{aligned}$$

$$\implies A \otimes_k B = k[T_1, \ldots, T_n, S_1, \ldots, S_r]/(f_1, \ldots, f_m, g_1, \ldots, g_s)$$

and

$$\mathrm{Sp}(A \otimes_k B) = V(f_1, \ldots, f_m) \times V(g_1, \ldots, g_s) \subseteq k^{n+r},$$

just as desired for a product!

- $\mathrm{Sp}(A \otimes_k B)$ also has the universal property: For $Z = \mathrm{Sp}(C)$, we have

$$\begin{aligned} \mathrm{Mor}(Z, \mathrm{Sp}(A \otimes_k B)) &\underset{\text{Hilbert}}{=} \mathrm{Hom}_{k\text{-alg}}(A \otimes_k B, C) \\ &\underset{-\otimes_k -}{=} \mathrm{Hom}_{k\text{-alg}}(A, C) \times \mathrm{Hom}_{k\text{-alg}}(B, C) \\ &\underset{\text{Hilbert}}{=} \mathrm{Mor}(Z, X) \times \mathrm{Mor}(Z, Y). \end{aligned}$$

$\implies \mathrm{Sp}(A \otimes_k B)$ has univ. prop of $X \times Y$!

Conclusion. If $G = \mathrm{Sp}(A)$, then $G \times G = \mathrm{Sp}(A \otimes_k A)$. Condition on m is that it is defined by polynomials, equivalently that it comes from a map $A \longrightarrow A \otimes_k A$, and that it gives a group structure to the set underlying G.

Example 1.20. The *additive group* $\mathbb{G}_a$.

$\mathbb{G}_a = (k, +)$ with variety structure $k = \mathrm{Sp}(k[t])$.

$$+ : k \times k \longrightarrow k, \qquad (x, y) \longmapsto x + y$$

is the polynomial map corresponding on rings to

$$\begin{aligned} k[x] \otimes k[y] &\xleftarrow{\varphi} k[t] \\ x \otimes 1 + 1 \otimes y &\longleftarrow\!\shortmid t. \end{aligned}$$

Example 1.21. The *multiplicative group* $\mathbb{G}_m$.

$\mathbb{G}_m = (k^\times, *)$ with variety structure $k^\times = \mathrm{Sp}(k[t, t^{-1}])$.

$$* : k^\times \times k^\times \longrightarrow k^\times, \qquad (x, y) \longmapsto x \cdot y$$

induced from

$$\begin{aligned} k[x^{\pm 1}] \otimes k[y^{\pm 1}] &\longleftarrow k[t^{\pm 1}] \\ x \otimes y &\longleftarrow\!\shortmid t. \end{aligned}$$

Remark 1.22. The above considered $k^\times$ as an open subset of $\mathbb{A}^1$. One can also treat a localization as a closed subset in an affine space of one dimension higher. For $\mathbb{G}_m$, this means identifying

$$\begin{aligned} k[t_1, t_2]/(t_1 t_2 - 1) &\xrightarrow{\sim} k[t, t^{-1}] \\ t_1, t_2 &\longmapsto t, t^{-1}. \end{aligned}$$

In these coordinates, multiplication is obviously given by polynomials:

$$\begin{array}{ccc} k^\times \times k^\times & \xrightarrow{m := *} & k^\times \\ \downarrow & & \downarrow \\ k^2 \times k^2 & \xrightarrow{\widetilde{m}} & k^2 \end{array} \qquad \begin{array}{ccc} (x, y) & \longmapsto & xy \\ \downarrow & & \downarrow \\ (x, x^{-1}, y, y^{-1}) & \longmapsto & (xy, (xy)^{-1}) \end{array}$$

$$\widetilde{m} : (x_1, x_2, y_1, y_2) \longmapsto (x_1 x_2, y_1 y_2).$$

Example 1.23. The *special linear* group SL_n.

$\mathrm{SL}_n = (\mathrm{SL}_n(k), *)$ with variety structure

$$\mathrm{SL}_n = \mathrm{Sp}\big(k[T_{11}, \ldots, T_{nn}]/(\det((T_{ij})_{i,j}) - 1)\big)$$

Here, $\det((T_{ij})_{i,j})$ is the determinant of the matrix with $T_{i,j}$ in the (i,j)-th entry. This is a polynomial in $k[T_{11}, \ldots, T_{nn}]$.

$$\begin{array}{ccc} \mathrm{SL}_n \times \mathrm{SL}_n & \xrightarrow{m := *} & \mathrm{SL}_n \\ \downarrow & & \downarrow \det = 1 \\ k^{n^2} \times k^{n^2} & \xrightarrow{\widetilde{m}} & k^{n^2} \end{array}$$

$$\widetilde{m} : ((x_{ij}), (y_{ij})) \longmapsto \left(\sum_{k=1}^{n} x_{ik} y_{kj} \right)_{i,j}$$

The morphism $\widetilde{m}$ preserves the determinant 1 condition and hence restricts to a multiplication morphism on SL_n.

Example 1.24. The *general linear group* GL_n

$\mathrm{GL}_n = (\mathrm{GL}_n(k), *)$ with variety structure

$$\mathrm{GL}_n = \mathrm{Sp}\big(k[T_{ij}, D]/(D \cdot \det(T_{ij}) - 1)\big) \subseteq k^{n^2} \times k.$$

The relation $D \cdot \det((T_{i,j})_{i,j}) - 1$ forces that D is the inverse of $\det((T_{ij})_{i,j})$, making the determinant invertible. This is the same as localizing at $\det((T_{i,j})_{i,j})$ as in Remark 1.22 before. We have

$$\begin{array}{ccc} \mathrm{GL}_n \times \mathrm{GL}_n & \longrightarrow & \mathrm{GL}_n \\ \downarrow & & \downarrow \\ (k^{n^2} \times k) \times (k^{n^2} \times k) & \longrightarrow & k^{n^2} \times k \end{array} \qquad \begin{array}{c} (x_{ij}) \\ \downarrow \\ ((x_{ij}), \det((x_{ij})_{i,j}) \end{array}$$

$$((x_{ij}), D_x, (y_{ij}), D_y) \longmapsto ((x_{ij}) \cdot (y_{jk}), D_x \cdot D_y).$$

Example 1.25. Let $H \in \mathrm{M}_n(k)$ be a non-degenerate symmetric matrix, i.e. $H^t = H$ and $\det(H) \neq 0$. The *special orthogonal group* defined by H is

$$\mathrm{SO}(H) = \{g \in \mathrm{SL}_n(k) \mid g^t H g = H\} \subseteq \mathrm{SL}_n.$$

Writing out the condition $g^t H g = H$ gives n^2 many quadratic polynomial equations $f_1, \ldots, f_{n^2}$ in the g_{ij}. Correspondingly, the variety structure on $\mathrm{SO}(H)$ is by realizing it as

$$\mathrm{SO}(H) = \mathrm{Sp}\big(k[g_{11}, \ldots, g_{nn}]/(f_1, \ldots, f_{n^2}, \det(g_{ij}) - 1)\big).$$

The group structure is matrix multiplication from SL_n.

Observation. We always assume that k is algebraically closed, so there exists a unique non-degenerate, n variable quadratic form up to isometry.

$\Longrightarrow$ After change of basis:

$$H = \begin{pmatrix} & \mathbb{1}_m \\ \mathbb{1}_m & \end{pmatrix} \quad n = 2m \quad \text{or} \quad H = \begin{pmatrix} 1 & & \\ & & \mathbb{1}_m \\ & \mathbb{1}_m & \end{pmatrix} \quad n = 2m+1.$$

Here $\mathbb{1}_m$ denotes the $m \times m$ identity matrix. For comparison, recall that over $\mathbb{R}$ a quadratic form is classified by its signature. Correspondingly, there would be several n variable orthogonal groups.

Example 1.26. The *symplectic group* Sp_{2m} over k is defined by

$$\mathrm{Sp}_{2m}(k) = \left\{ g \in \mathrm{M}_{2m}(k) \;\middle|\; g^t \begin{pmatrix} & \mathbb{1}_m \\ -\mathbb{1}_m & \end{pmatrix} g = \begin{pmatrix} & \mathbb{1}_m \\ -\mathbb{1}_m & \end{pmatrix} \right\} \subseteq k^{4m^2}.$$

It is described by $4m^2$ polynomial equations with matrix multiplication just as the special orthogonal group before.

Remark 1.27. $\mathrm{SL}_n, \mathrm{SO}_{2m}, \mathrm{SO}_{2m+1}$ and Sp_{2m} are called the *classical groups*.

2. Topological properties (9/23/2026)

k always denotes an algebraically closed field of characteristic 0.

2.1. **Examples.** Our first goal today is to see more examples of linear algebraic groups and some related easy definitions. Recall that a linear algebraic group G over k is an affine k-variety G together with a multiplication morphism $m : G \times G \to G$ giving its set of points a group structure.

In other words, G can be written as the vanishing locus of a family of polynomials equations $G \subseteq k^n$ in some k^n such that m is induced from a (non-unique) polynomial map $\widetilde{m}$:

$$\begin{array}{ccc} G \times G & \xrightarrow{\quad m \quad} & G \\ \big\downarrow & & \big\downarrow \\ k^n \times k^n & \xrightarrow{\quad \widetilde{m} \quad} & k^n \end{array}$$

Note. Our theory of varieties is built around *reduced* finite type k-algebras. In scheme theory language, we are considering reduced finite-type k-schemes.

Example 2.1. Let H be a symmetric $n \times n$ matrix with $\det(H) \in k^\times$. The *orthogonal group* is

$$O(H) := \left\{x \in \mathrm{M}_n(k) : x^t H x = H\right\}$$

with variety structure as a closed subvariety of $\mathrm{M}_n(k) \cong \mathbb{A}^{n^2}$. Then, by our conventions, the orthogonal group has the variety structure $O(H) = \mathrm{Sp}(A_{red})$ with

$$A = k[x_{11}, x_{12}, ..., x_{nn}]/(x^t H x - H)$$

where the ideal $(x^t H x - H) \subseteq k[x_{11}, ..., x_{nn}]$ has n^2 many partially redundant equations. It turns out, however, that A is already reduced.

The same method and remark apply to the symplectic group

$$\mathrm{Sp}_{2m} := \left\{x \in \mathrm{M}_{2m}(k) : x^t \begin{pmatrix} & \mathbb{1}_m \\ -\mathbb{1}_m & \end{pmatrix} x = \begin{pmatrix} & \mathbb{1}_m \\ -\mathbb{1}_m & \end{pmatrix}\right\}$$

The deeper reason for this phenomenon is that *every* finite type group scheme over a field of characteristic 0 is reduced (even smooth). This is the reason for why we work over a field of characteristic 0.

Example 2.2. Let $H = \left(\begin{smallmatrix} & 1 \\ 1 & \end{smallmatrix}\right)$. Then

$$\mathrm{SO}(H) = \left\{x \in \mathrm{M}_2(k) : \det(x) = 1, x^t H x = H\right\}.$$

is regarded as $\mathrm{Sp}(A)$ where $A = k[x_{11}, x_{12}, x_{21}, x_{22}]/I$, here

$$I := (\underbrace{x_{11}x_{22} - x_{12}x_{21} - 1}_{\det(x)=1}, \quad x_{11}x_{22} + x_{12}x_{21} - 1, \quad 2x_{11}x_{21}, \quad 2x_{22}x_{12}).$$

The later three generators of this ideal come from making explicit the $x^t H \overset{!}{=} H$ entry by entry:

$$\begin{aligned}\begin{pmatrix} x_{11} & x_{21} \\ x_{12} & x_{22} \end{pmatrix} \begin{pmatrix} & 1 \\ 1 & \end{pmatrix} \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix} &= \begin{pmatrix} x_{21} & x_{11} \\ x_{22} & x_{12} \end{pmatrix} \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix} \\ &= \begin{pmatrix} 2x_{11}x_{21} & x_{12}x_{21} + x_{11}x_{22} \\ x_{12}x_{21} + x_{11}x_{22} & 2x_{22}x_{12} \end{pmatrix} \\ &\overset{!}{=} \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}.\end{aligned}$$

Then, we have the following relations in A:

$$x_{12}x_{21} = -x_{12}x_{21} \implies x_{12}x_{21} = 0 \implies x_{11}x_{22} = 1.$$

In particular, the elements x_{11}, x_{22} are invertible, so $x_{12} = x_{21} = 0$ in A. Hence,

$$A \cong k[x_{11}, x_{22}]/(x_{11}x_{22} - 1),$$

so we get the identification

$$\mathrm{Sp}(A) = \left\{\begin{pmatrix} \lambda & \\ & \lambda^{-1} \end{pmatrix} : \lambda \in k^\times\right\} \subseteq \mathrm{M}_2(k)$$

which is isomorphic to $\mathbb{G}_m$.

Definition 2.3 (Homomorphisms of LAGs)**.** Let G, H be LAGs$/k$. A *linear algebraic group homomorphism* $f : G \to H$ is a map that is both a group homorphism and morphism of algebraic varieties.

Example 2.4. The determinant defines a homomorphism $\det : \mathrm{GL}_n \to \mathbb{G}_m$,

$$\begin{array}{ccc} \mathrm{GL}_n & \xrightarrow{f := \det} & \mathbb{G}_m \\ \downarrow & & \downarrow \\ k^{n^2} \times k & \xrightarrow{\widetilde{f}} & k \times k \end{array} \qquad \begin{array}{ccc} (x_{ij}) & \longmapsto & \det(x_{ij}) \\ \downarrow & & \downarrow \\ (x_{ij}, \det(x_{ij})^{-1}) & \longmapsto & (\det(x_{ij}), \det(x_{ij})^{-1}). \end{array}$$

The vertical embeddings here are the same as in Examples 1.21 and 1.24. The map corresponding to det on coordinate rings is

$$\begin{aligned} k[x_{ij}, D]/(\det(x_{ij}) \cdot D - 1) &\longleftarrow k[t, t^{-1}] \\ \det(x_{ij}) &\longleftarrow\!\shortmid \quad t \end{aligned}$$

Example 2.5. Let H be a comutative LAG over k, with multiplication $m_H : H \times H \to H$. Then we can multiply $f_1, f_2 : G \to H$ by

$$(f_1 \cdot f_2)(g) = f_1(g) \cdot f_2(g).$$

In this way, $\mathrm{Hom}_{LAG}(G, H)$ becomes a group. Moreover, $\mathrm{End}_{LAG}(H, H)$ is a ring with addition coming from m_H and multiplication being composition.

Exercise. Show that there is an isomorphism $k \xrightarrow{\sim} \mathrm{End}_{LAG}(\mathbb{G}_a)$ given by

$$\alpha \longmapsto [k \mapsto k, x \mapsto \alpha x].$$

In the same way, show that

$$\mathrm{M}_m(k) \xrightarrow{\sim} \mathrm{End}_{LAG}(\mathbb{G}_a^m).$$

Proposition 2.6. *There is an isomorphism of rings*

$$\mathbb{Z} \xrightarrow{\sim} \mathrm{End}_{LAG}(\mathbb{G}_m), \qquad m \longmapsto [x \mapsto x^m].$$

Proof. The variety morphisms $f : \mathbb{G}_m \to \mathbb{G}_m$ correspond to the k-algebra homomorphism $f^* : k[t^{\pm 1}] \to k[t^{\pm 1}]$. The units of $k[t^{\pm 1}]$ are in form of $\alpha \cdot t^m$, for $\alpha \in k^\times$ and $m \in \mathbb{Z}$. Thus

$$f^*(t) = \alpha t^m$$

for unique $\alpha \in k^\times$ and $m \in \mathbb{Z}$, meaning that $f(x) = \alpha x^m$. The group homomorphism condition is equivalent to

$$\begin{aligned} f(xy) = f(x)f(y) &\iff \alpha x^m y^m = \alpha^2 x^m y^m \\ &\iff \alpha = \alpha^2 \\ &\iff \alpha = 1. \end{aligned}$$

Hence $\mathrm{End}_{LAG}(\mathbb{G}_m) = \{x \mapsto x^m \mid m \in \mathbb{Z}\}$. □

Example 2.7 (Inner automorphisms)**.** Let G be a LAG over k and fix $x \in G$. Then

$$G \xrightarrow{\sim} G, \qquad g \longmapsto xgx^{-1},$$

is an automorphism of linear algebraic groups. It is a morphism because it is the composition of

$$G \cong \{x\} \times G \times \{x^{-1}\} \hookrightarrow G \times G \times G \xrightarrow{m \circ (\mathrm{id} \times m)} G.$$

Definition 2.8 (Subgroups of a LAG)**.** Let $X = \mathrm{Sp}(A)$ be an affine variety. A *closed subvariety* of X is a closed subset $Z \subseteq X$, endowed with the variety structure $\mathrm{Sp}(A/I(Z))$.

For a linear algebraic group G, a *subgroup of* G always means a closed subvariety $H \subseteq G$ that is also a subgroup.

Example 2.9. The subgroup $\mathrm{SL}_n \subseteq \mathrm{GL}_n$ is the closed subvariety defined by $\det = 1$:

$$V(\det(x_{ij}) - 1) \subseteq \mathrm{Sp}\big(k[x_{ij}, D]/(\det(x_{ij})D - 1)\big).$$

Example 2.10. The subgroup

$$\left\{ \begin{pmatrix} x_{11} & x_{12} \\ 0 & 1 \end{pmatrix} \;\middle|\; x_{11} \in k^\times,\ x_{12} \in k \right\} \subseteq \mathrm{GL}_2$$

is the closed subvariety defined by $x_{21} = 0$ and $x_{22} = 1$.

Example 2.11. If $H_1, H_2 \subseteq G$ are closed subgroups, then $H_1 \cap H_2$ is a closed subgroup. For example,

$$\left\{ \begin{pmatrix} x_{11} & x_{12} \\ 0 & x_{22} \end{pmatrix} \right\} \cap \mathrm{SL}_2 = \left\{ \begin{pmatrix} x_{11} & x_{12} \\ 0 & x_{11}^{-1} \end{pmatrix} \right\},$$

where the first set is viewed inside GL_2. The intersection is closed and can be defined by $x_{21} = 0$ and $x_{11}x_{22} = 1$.

Example 2.12 (Kernels are closed subgroups)**.** Let $f : G \to H$ be a homomorphism of linear algebraic groups. Since f is continuous and $\{e_H\} \subseteq H$ is closed,

$$\ker(f) = f^{-1}(e_H)$$

is a closed subgroup of G.

We generalize the definition of kernels using fiber products. First recall the definition:

Definition 2.13. Given maps of sets $X \xrightarrow{f} S \xleftarrow{g} Y$, their *fibre product* is

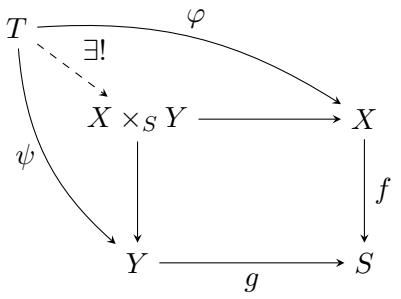
$$X \times_S Y = \{(x, y) \in X \times Y \mid f(x) = g(y)\}.$$

In other words, for every $s \in S$, we take the product of the fibres $f^{-1}(s) \times g^{-1}(s) \subseteq X \times Y$.

Universal property. The fiber product has the following property: For every set T,

$$\mathrm{Map}(T, X \times_S Y) = \{(\varphi, \psi) \in \mathrm{Map}(T, X) \times \mathrm{Map}(T, Y) \mid f \circ \varphi = g \circ \psi\}.$$

This is better memorized by the following diagram, where the condition is that the outer square is commutative:



Instead of working with just sets, this whole construction also works in the category of affine varieties over k. Consider

$$S = \mathrm{Sp}(A), \qquad X = \mathrm{Sp}(B), \qquad Y = \mathrm{Sp}(C),$$

with morphisms $f\colon X \to S$ and $g\colon Y \to S$.

Proposition 2.14. *The affine variety* $\mathrm{Sp}\big((B \otimes_A C)_{\mathrm{red}}\big)$ *is a fibre product of* X *and* Y *over* S.

Proof. Let D be any reduced k-algebra of finite type. Then

$$\begin{aligned}
&\mathrm{Mor}_{k\text{-var}}\big(\mathrm{Sp}(D), \mathrm{Sp}((B \otimes_A C)_{\mathrm{red}})\big) \\
&\quad = \quad \mathrm{Mor}_{k\text{-alg}}\big((B \otimes_A C)_{\mathrm{red}}, D\big) \\
&\quad \overset{D \text{ reduced}}{=} \mathrm{Mor}_{k\text{-alg}}(B \otimes_A C, D) \\
&\quad = \quad \big\{\ (\varphi^*\colon B \to D,\ \psi^*\colon C \to D) \bigm| \varphi^* \circ f^* = \psi^* \circ g^*\ \big\} \\
&\quad = \quad \big\{\ (\varphi\colon \mathrm{Sp}(D) \to X,\ \psi\colon \mathrm{Sp}(D) \to Y) \bigm| f \circ \varphi = g \circ \psi\ \big\}.
\end{aligned}$$

□

Now assume all the morphisms involved are homomorphisms of linear algebraic groups:

$$\begin{array}{ccc} & & G_2 \\ & & \big\downarrow f_2 \\ G_1 & \xrightarrow[f_1]{} & H \end{array}$$

Then

$$G = G_1 \times_H G_2 \subseteq G_1 \times G_2$$

is a LAG and a closed subgroup of $G_1 \times G_2$. For $G_1 = \{e_H\} \hookrightarrow H$, this recovers $\ker(f_2)$.

Example 2.15. The fibre-product square

$$\begin{array}{ccc} G & \longrightarrow & \mathrm{GL}_n \\ \big\downarrow & \square & \big\downarrow \det \\ \mathrm{GL}_n & \xrightarrow[\det]{} & \mathbb{G}_m \end{array}$$

gives

$$G = \big\{(g_1, g_2) \in \mathrm{GL}_n \times \mathrm{GL}_n \mid \det(g_1) = \det(g_2)\big\}.$$

2.2. **Topological properties.** Reference: [1, §§1.1–1.3].

In this section, we will study the three topological properties of being quasi-compact, irreducible, and connected.

2.2.1. *Quasi-compactness.*

Definition 2.16. A topological space X is called *quasi-compact* if every open covering $X = \bigcup_{i \in I} U_i$ has a finite subcover.

Proposition 2.17. *For every reduced k-algebra A of finite type, $X = \mathrm{Sp}(A)$ is quasi-compact.*

Proof. Let $X = \bigcup_{i \in I} U_i$ be an open covering of X. We first cover each U_i by principal open subsets $D(f_{ij})$, $j \in J_i$, using that these form a basis of the topology. Clearly, if we can even find finitely many $D(f_{i_1,j_1}), \ldots, D(f_{i_r,j_r})$ covering X, then larger $U_{i_1}, \ldots, U_{i_r}$ also cover X. So we may assume all $U_i = D(f_i)$ principal to begin with.

Recall that $D(f) \subseteq \mathrm{Sp}(A)$ is the subset of maximal ideals $\mathfrak{m} \subseteq A$ not containing f. So $X = \bigcup_{i \in I} D(f_i)$ means that no maximal ideal of A contains all f_i. In other words, the ideal $(f_i)_{i \in I}$ generated by the f_i is not contained in any maximal ideal. This means it equals the unit ideal. So there exist indices $i_1, \ldots, i_r$ and a way to write

$$1 = h_1 f_{i_1} + \ldots + h_r f_{i_r}.$$

Then $X = D(f_{i_1}) \cup \ldots \cup D(f_{i_r})$ is the desired finite subcovering. □

Remark 2.18. This proof did not use any Noetherian property, but the following exercise does.

Exercise. Let $U \subseteq X$ be any open subset. Using that A is noetherian, show that $Z = X \setminus U$ is the vanishing set of finitely many $f_1, \ldots, f_r \in A$. Deduce that the $D(f_i)$ form an open covering of U. Deduce further that U is quasi-compact.

2.2.2. *Irreducibility.*

Definition 2.19. A topological space X is *irreducible* for all closed $Z_1, Z_2 \subseteq X$ with $X = Z_1 \cup Z_2$, $Z_1 = X$ or $Z_2 = X$.

A subset $Z \subseteq X$ is irreducible if it is irreducible with respect to the subspace topology. This means that whenever $Z \subseteq Z_1 \cup Z_2$ for two closed subsets $Z_1, Z_2 \subseteq X$, then already $Z \subseteq Z_1$ or $Z \subseteq Z_2$.

Proposition 2.20. *Let $X = \mathrm{Sp}(A)$. There is a bijection*

$$\begin{aligned} \{\textit{prime ideals } \mathfrak{p} \subset A\} &\overset{1:1}{\longleftrightarrow} \{\textit{nonempty irreducible closed } Z \subseteq X\}, \\ \mathfrak{p} &\longmapsto V(\mathfrak{p}), \\ I(Z) &\longmapsfrom Z. \end{aligned}$$

Proof. Let $\varnothing \neq Z \subseteq X$ be closed and irreducible, and suppose $fg \in I(Z)$. Then

$$Z \subseteq V(f) \cup V(g).$$

By irreducibility, $Z \subseteq V(f)$ or $Z \subseteq V(g)$. Thus $f \in I(Z)$ or $g \in I(Z)$, so $I(Z)$ is a prime ideal.

Conversely, let $\mathfrak{p}$ be a prime ideal, and suppose that $V(\mathfrak{p}) \subseteq Z_1 \cup Z_2$ for closed subsets Z_1, Z_2. Assume that $V(\mathfrak{p}) \nsubseteq Z_1$ and $V(\mathfrak{p}) \nsubseteq Z_2$. Then by Hilbert's Nullstellensatz, there are functions

$$f_i \in I(Z_i) \smallsetminus \mathfrak{p} \qquad (i = 1, 2),$$

The product $f_1 f_2$ vanishes on $Z_1 \cup Z_2$, hence on $V(\mathfrak{p})$. Again by Hilbert's Nullstellensatz, $f_1 f_2 \in \mathfrak{p}$. Since $\mathfrak{p}$ is prime, this implies $f_1 \in \mathfrak{p}$ or $f_2 \in \mathfrak{p}$, a contradiction. □

Proposition 2.21. *Any closed subset $Z \subseteq X = \mathrm{Sp}(A)$ is a finite union of irreducible closed subsets.*

Proof. Suppose there is a closed subset $Z_0 \subseteq X$ that cannot be written as a finite union of irreducible closed subsets. In particular, Z_0 is not irreducible, so

$$Z_0 = Z_1 \cup Z_1', \qquad Z_1 \subsetneq Z_0, \quad Z_1' \subsetneq Z_0,$$

and at least one of Z_1 and Z_1' cannot be written as a finite union of irreducible closed subsets. Say this is Z_1. Continuing in this way, we obtain a strictly descending chain

$$Z_0 \supsetneq Z_1 \supsetneq Z_2 \supsetneq \cdots.$$

By Hilbert's Nullstellensatz, this corresponds to a *strictly* ascending chain of ideals

$$\mathfrak{a}_0 \subsetneq \mathfrak{a}_1 \subsetneq \mathfrak{a}_2 \subsetneq \cdots, \qquad \mathfrak{a}_i = I(Z_i).$$

The strictness of the inclusions is essential here and comes from the bijectivity part of Hilbert's Nullstellensatz. But A is Noetherian, so this chain must stabilize, contradicting our original assumptions. □

Corollary 2.22. *Any closed $Z \subseteq \mathrm{Sp}(A)$ can be written as a finite union*

$$Z = Z_1 \cup \cdots \cup Z_r, \qquad Z_i \not\subseteq Z_j \quad \text{for } i \neq j,$$

of closed irreducible subsets. The Z_i are called the irreducible components *of Z. They are in bijection as*

$$\{\text{irreducible components of } Z\} \xleftrightarrow{1:1} \left\{\begin{matrix}\text{minimal prime ideals} \\ \text{of } A \text{ containing } I(Z)\end{matrix}\right\}. \tag{2.1}$$

The corollary of course also applies to $\mathrm{Sp}(A)$ itself, and in fact is equivalent to this case by $Z = \mathrm{Sp}(A/I(Z))$.

2.2.3. *Connectedness.*

Definition 2.23. A topological space X is *connected* if every subset $U \subseteq X$ that is both open and closed is either $\varnothing$ or X. Equivalently, we cannot write

$$X = U_1 \sqcup U_2$$

as a disjoint union of two nonempty open subsets.

Lemma 2.24. *Let $X = \mathrm{Sp}(A)$. There is a bijection*

$$\{U \subseteq X \mid U \text{ is open and closed}\} \longleftrightarrow \{e \in A \mid e^2 = e\},$$

between open and closed subsets and idempotents in A, given by

$$\begin{array}{rcl} U & \longmapsto & \mathbb{1}_U \in \mathscr{O}(X), \\ D(e) & \longmapsfrom & e. \end{array}$$

Here $\mathbb{1}_U$ denotes the indicator function of U.

Proof. We first show that $\mathbb{1}_U$ is regular, i.e. that it lies in $\mathscr{O}(X)$. Since U and U^c are both open, we can cover them by principal opens as

$$U = \bigcup_i D(f_i), \qquad U^c = \bigcup_j D(g_j).$$

Then

$$\mathbb{1}_U|_{D(f_i)} = 1 \in A[f_i^{-1}], \qquad \mathbb{1}_U|_{D(g_j)} = 0 \in A[g_j^{-1}].$$

Thus, taking our definition of regularity (Definition 1.13), we get that $\mathbb{1}_U$ is regular as claimed. By the mutual quasi-inverse part of the strong form of Hilbert's Nullstellensatz (Theorem 1.18), this means $\mathbb{1}_U$ equals (the function defined by) some $e \in A$. The map $U \mapsto \mathbb{1}_U$ is clearly injective and $\mathbb{1}_U^2 = \mathbb{1}_U$.

Conversely, if $e^2 = e$, then $e(x)^2 = e(x)$ for every $x \in X$. Thus $e(x) \in \{0, 1\}$ and

$$X = D(e) \sqcup D(1 - e).$$

This shows that $D(e)$ is both open and closed, so the map from right to left is well defined. The two maps are clearly mutually inverses. □

Exercise. (a) Let $X_1, X_2 \subseteq X$ be two subsets of a topological space which are connected with respect to the subspace topology. Show that if $X_1 \cap X_2 \neq \emptyset$, then $X_1 \cup X_2$ is connected as well.

(b) Specialize to $X = \mathrm{Sp}(A)$ and its irreducible components $X_1, \ldots, X_r$. Show that the X_i can be uniquely grouped by $\{1, \ldots, n\} = I_1 \sqcup \ldots \sqcup I_k$ such that each union $\bigcup_{i \in I_j} X_i$ is connected, open and closed in X. These unions are called the *connected components* of X.

The exercise in particular proves the following proposition:

Proposition 2.25. *Let A be a finite type k-algebra. Then A has only finitely many idempotents.*

Algebraically, we can uniquely up to isomorphism decompose

$$A \cong A_1 \times \cdots \times A_k$$

as a product such that $0, 1 \in A_i$ are the only idempotents. Geometrically, this corresponds to a disjoint union decomposition

$$\operatorname{Sp}(A) = \operatorname{Sp}(A_1) \sqcup \cdots \sqcup \operatorname{Sp}(A_k)$$

with each $\operatorname{Sp}(A_i)$ is open, closed, and connected. Each $\operatorname{Sp}(A_i)$ corresponds to a union $\bigcup_{i \in I_j} X_i$ of irreducible components in the terminology of the above exercise.

Example 2.26. If A is an integral domain, then $\operatorname{Sp}(A)$ is connected, because the only solutions of $e(e-1) = 0$ are $e = 0$ and $e = 1$.

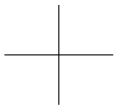
Remark 2.27. The real picture can sometimes be deceptive:

$$\operatorname{Sp}(k[t^{\pm 1}]) \quad = \quad$$

is connected as $k[t, t^{-1}]$ is an integral domain. It is better to think of $\mathbb{C} \smallsetminus \{0\}$ which is connected.

Example 2.28. Let $A = k[x, y]/(xy)$. Geometrically,

$$\operatorname{Sp}(A) \quad = \quad$$



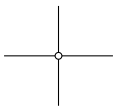
$$\subseteq \ k^2.$$

There are two irreducible components:

$$\operatorname{Sp}(A) = V(x) \cup V(y).$$

Consider $x + y \in A$. It vanishes only at $(0, 0)$, so

$$\operatorname{Sp}\big(A[(x+y)^{-1}]\big) \quad = \quad$$



.

In $A[(x+y)^{-1}]$, we calculate

$$\left(\frac{x}{x+y}\right)^2 = \frac{x^2}{x^2+2xy+y^2} = \frac{x^2}{x^2+y^2} = \frac{x}{x+y},$$

because $x(x^2+y^2) = x^2(x+y)$ in A (clear denominators). Therefore, $x/(x+y)$ and similarly $y/(x+y)$ are idempotents in $A[(x+y)^{-1}]$. This is clear geometrically, because $x/(x+y)$ is 1 on the axis $\{y = 0\}$ and 0 on the axis $\{x = 0\}$.

One checks that this in fact gives an isomorphism

$$A[(x+y)^{-1}] \cong k[x, x^{-1}] \times k[y, y^{-1}],$$

which describes the two connected components of $\operatorname{Sp}(A[(x+y)^{-1}]$ in line with Proposition 2.25.

2.3. **Connectedness for LAGs.** Reference: [1, §7.3].

We now want to study the topological properties of linear algebraic groups. First, we need a lemma.

Lemma 2.29. *Let X and Y be irreducible affine k-varieties. Then $X \times Y$ is again irreducible.*

Proof. Suppose that $X \times Y = Z_1 \cup Z_2$ is a union of two closed subsets. For every $x \in X$, the subset $\{x\} \times Y$ is closed, and by irreducibility of Y we have

$$\{x\} \times Y \subseteq Z_1 \quad \text{or} \quad \{x\} \times Y \subseteq Z_2.$$

Define

$$X_i = \{x \in X \mid \{x\} \times Y \subseteq Z_i\} \qquad (i = 1, 2).$$

Then $X = X_1 \cup X_2$ by the argument just given. We may also write X_i as the (usually infinite) intersection

$$X_i = \bigcap_{y \in Y} (X \times \{y\} \cap Z_i)\,.$$

This shows that X_i is closed. Since X is irreducible, either $X = X_1$ or $X = X_2$. Therefore, either $X \times Y = Z_1$ or $X \times Y = Z_2$. □

Lemma 2.30. *There is a unique irreducible component $G^\circ \subseteq G$ containing the identity e.*

Proof. Let $X_1, \ldots, X_r \subseteq G$ be the irreducible components containing e. By Lemma 2.29, $X_1 \times \cdots \times X_r$ is irreducible. Consider the multiplication morphism

$$m^{(r)} : G^r \longrightarrow G.$$

The image of an irreducible topological space under a continuous map is irreducible (exercise). Hence

$$Y := m^{(r)}(X_1 \times \cdots \times X_r) \subseteq G$$

is irreducible. Since $e \in Y$, we have $Y \subseteq X_i$ for some i. But $X_j \subseteq Y$ for every j and $X_i \not\subseteq X_j$ whenever $i \neq j$. So the only possibility is $r = 1$, proving the lemma. □

Proposition 2.31. *The identity component $G^\circ \subseteq G$ is an open normal subgroup. Its cosets are the irreducible components of G, which are simultaneously the connected components of G.*

Proof. • By Lemma 2.29, $G^\circ \times G^\circ$ is irreducible, so $m(G^\circ \times G^\circ)$ is contained in an irreducible component of G. Since $e \in m(G^\circ \times G^\circ)$, this compenent has to be G°. Moreover, $m(g, e) = g$ for every $g \in G^\circ$, and therefore

$$m(G^\circ \times G^\circ) = G^\circ.$$

Thus multiplication restricts to a morphism

$$m : G^\circ \times G^\circ \longrightarrow G^\circ.$$

• For every $x \in G$, left translation

$$L_x : G \longrightarrow G, \qquad g \longmapsto xg,$$

is a morphism because it is the composition of two morphism:

$$\{x\} \times G \longrightarrow G \times G \xrightarrow{m} G.$$

It is even an isomorphism with inverse $g \mapsto x^{-1}g$. For each $x \in G^\circ$, the translate $x^{-1}G^\circ$ is therefore again an irreducible component of G. It contains e because $x \in G^\circ$, so $x^{-1}G^\circ = G^\circ$. In particular, $x^{-1} = x^{-1} \cdot e \in G^\circ$ which means that G° is stable under $x \mapsto x^{-1}$ and hence a closed subgroup.

• Since G is an affine variety, it has finitely many irreducible components. Write

$$G = G^\circ \cup \bigcup_{i=1}^{r} X_i, \qquad \text{each } X_i \neq G^\circ, \quad X_i \neq X_j \text{ for } i \neq j.$$

If $x \in X_i$, translation by x is an isomorphism of G. In particular, $x^{-1}X_i$ is an irreducible component containing e, so $x^{-1}X_i = G^\circ$. Thus $X_i = xG^\circ$, and all irreducible components are cosets of G°.

• Different cosets are disjoint, i.e. $xG^\circ \cap yG^\circ = \varnothing$ whenever $xG^\circ \neq yG^\circ$. So for each xG°, we can write

$$xG^\circ = G \smallsetminus \bigcup_{yG^\circ \neq xG^\circ} yG^\circ.$$

The union on the right is a finite union of closed subsets, hence closed. Thus the complement xG° is open.

• Finally, for normality, consider any $x \in G$. Conjugation

$$c_x : G \longrightarrow G, \qquad g \longmapsto xgx^{-1},$$

is an automorphism with inverse $c_{x^{-1}}$ (Example 2.7). Hence $xG^\circ x^{-1}$ is an irreducible component again. Since it contains e, it equals G°. □

We end this section with some examples.

Example 2.32. The general linear group GL_n is the principal open subset $D(\det(x_{ij}))$ in $\mathrm{M}_n(k)$. In particular, its coordinate ring

$$k[\mathrm{GL}_n] := \mathscr{O}_{\mathrm{GL}_n}(\mathrm{GL}_n)$$

is the localization $k[x_{11}, \ldots, x_{nn}, \det(x_{ij})^{-1}]$ and hence an integral domain. It follows that GL_n is irreducibe and connected, see (2.1). Therefore, $\mathrm{GL}_n^\circ = \mathrm{GL}_n$.

Example 2.33. Forgetting about the group structure on GL_n, i.e. viewing it solely as an affine variety, there exists an isomorphism

$$\mathrm{GL}_n \xleftarrow{\sim} \mathbb{G}_m \times \mathrm{SL}_n$$
$$\begin{pmatrix} \lambda & \\ & \mathbb{1}_{n-1} \end{pmatrix} \cdot (x_{ij}) \longleftarrow\!\shortmid (\lambda, (x_{ij})).$$

In general, a product of (non)-irreducible affine varieties is (non)-irreducible, hence SL_n is irreducible. That is, $\mathrm{SL}_n^\circ = \mathrm{SL}_n$.

Example 2.34. Given $k \geq 1$, consider the homomorphism

$$\varphi : \mathrm{GL}_n \longrightarrow \mathbb{G}_m, \qquad x \longmapsto \det(x)^k.$$

Its kernel is

$$\ker(\varphi) = \{x \in \mathrm{GL}_n \mid \det(x) \in \mu_k\}$$

where $\mu_k := \{1, \zeta, \ldots, \zeta^{k-1}\}$ is group of k-th roots of unity. By the previous example, its identity component is $\ker^\circ(\varphi) = \mathrm{SL}_n$ and $\ker(\varphi)/\ker(\varphi)^\circ \xrightarrow{\sim} \mu_k$.

Example 2.35. Consider a non-degenerate symmetric form $H \in \mathrm{Sym}_n(k)$ and the determinant homomorphism

$$\det : \mathrm{O}(H) \longrightarrow\!\!\!\!\!\rightarrow \{\pm 1\}.$$

Its kernel is $\mathrm{SO}(H)$, hence

$$\mathrm{O}(H)/\mathrm{SO}(H) \xrightarrow{\sim} \{\pm 1\}.$$

We see that $\mathrm{O}(H)^\circ \subseteq \mathrm{SO}(H)$, so $\mathrm{O}(H)$ is not connected.

Non-obvious fact: $\mathrm{SO}(H)$ is connected. Hence $O(H)^\circ = \mathrm{SO}(H)$.

In the same way, Sp_{2m} is connected.

3. Embedding into GL_n and images of morphisms (9/30/2026)

Let $\varphi : G \to H$ be a homomorphism of LAGs. We saw as an easy example last week that $\ker(\varphi) \subseteq G$ is a closed subgroup and, in particular, naturally again an LAG (Example 2.12). Today, our goal is to prove that also the image $\mathrm{Im}(\varphi) \subseteq H$ is a closed subgroup, which is at a whole different difficulty level. We need two new tools as inputs:

(1) We will show that for any LAG G, the map $G \to G$, $x \mapsto x^{-1}$ is a morphism of varieties. We will obtain this as a corollary to a fundamental theorem of the theory: Every LAG can be embedded as a closed subgroup into some GL_n.

(2) We will prove that images of variety morphisms are *constructible*. This is a basic result from algebraic geometry which we will discuss in the middle of the lecture.

Remark 3.1. Property (1) has to be imposed as axiom when working with potentially non-reduced group objects such as group schemes or not geometrically reduced varieties. It is also required from the beginning in [1, §7.1].

In our course, we started from slightly weaker axioms. We only assumed that G is an affine variety and that $m : G \times G \to G$ is a morphism of varieties that induces a group structure on G viewed as set. So, for example, we currently do not even know that $x \mapsto x^{-1}$ is a continuous map.

3.1. Embedding into GL_n.

Reference: [1, §8.6].

Theorem 3.2. *Let G/k be a LAG. There exist $n \in \mathbb{N}$ and an embedding as a closed subgroup*

$$G \overset{\varphi}{\hookrightarrow} \mathrm{GL}_n.$$

Here, by *embedding* we more precisely mean a closed immersion of affine varieties (i.e. taking a quotient on coordinate rings).

Corollary 3.3. *The inversion map $i : G \to G$, $x \mapsto x^{-1}$ is a variety automorphism.*

Proof. Choose an embedding $\varphi : G \hookrightarrow \mathrm{GL}_n$ as in Theorem 3.2. The inversion map

$$i : \mathrm{GL}_n \longrightarrow \mathrm{GL}_n, \qquad x \longmapsto x^{-1}$$

is a morphism because it is given by the polynomial formula $x^{-1} = \det(x)^{-1} \cdot x^{\mathrm{ad}}$ where x^{ad} is the adjunct matrix. Inversion on GL_n preserves the closed subset $\varphi(G) \subset \mathrm{GL}_n$ because G is a group and φ a homomorphism, so it induces an automorphism on the closed subvariety $\varphi(G)$. Since $i : G \xrightarrow{\sim} \varphi(G)$ by what it means to be a closed embedding, inversion restricts to an automorphism $i : G \longrightarrow G$ which is just $x \mapsto x^{-1}$. □

Goal. Prove Theorem 3.2 for a given LAG (G, m). Recall our three axioms:

- G is an affine k-variety
- $m : G \times G \longrightarrow G$ is a morphism of k-varieties
- m induces group structure on G viewed as set.

Translation. Recall that $k[G]$ is the ring consisting of the regular functions $f : G \to k$. For $x \in G$, we define the left x-translate of f by

$$(\lambda_x f)(y) := f(x^{-1}g).$$

We choose this convention because $\lambda_x(\lambda_y(f)) = \lambda_{xy}(f)$. That is, $x \mapsto \lambda_x$ is a left representation of G on $\mathrm{Map}(G, k)$.

Claim 0. For $x \in G$ and $f \in k[G]$, the translate $\lambda_x(f)$ again lies in $k[G]$.

Proof. Consider the composition

$$\psi_x : \{x^{-1}\} \times G \hookrightarrow G \times G \xrightarrow{m} G.$$

This is a composition of two morphisms, hence a morphism. So the pullback $\lambda_x(f) = f \circ \psi_x$ is again a regular function on G. □

Claim 1. Let $x \in G$ and $f \in k[G]$. There exists a finite dimensional k-subvector space $F \subseteq k[G]$ containing f such that F is stable under left translation by all $x \in G$. That is, for all functions $h \in F$ and $x \in G$, $\lambda_x(h)$ again lies in F.

Proof. Consider the pullback $m^*(f) = f \circ m$. It is a regular function on $G \times G$, i.e. lies in $k[G \times G] = k[G] \otimes_k k[G]$. Explicitly,

$$f(x, y) = f(xy).$$

Any element of $k[G] \otimes_k k[G]$ is a finite sum of elementary tensors, so there exist $a_1, \ldots, a_r$, $h_1, \ldots, h_r \in k[G]$ such that for all $x, y \in G$,

$$f(xy) = \sum_{i=1}^{r} a_i(x) \otimes h_i(y).$$

Then, for every $x \in G$, we get that

$$\lambda_x(f) = \sum_{i=1}^{r} a_i(x^{-1}) h_i$$

is a linear combination of $h_1, \ldots, h_r$. So the k-vector space $F_f := \operatorname{span}_k\{\lambda_x(f)\}_{x \in G}$ spanned by all left translates

- is finite-dimensional because it is contained in $\operatorname{span}_k\{h_1, \ldots, h_r\}$,
- contains $f = \lambda_e(f)$ and is stable under left translation because $\lambda_z(\lambda_x(f)) = \lambda_{zx}(f)$.

That is, F_f fulfilles all requirements of Claim 1. □

Claim 2. Assume $F \subseteq k[G]$ is a (finite or infinite-dimensional) k-subvector space that is stable under all left translations. Then $f \circ m \in k[G] \otimes_k F$ for all $f \in F$.

Proof. Choose a basis $\{f_i\}_{i \in I}$ for F. Complete it to a basis $\{f_i\}_{i \in I} \sqcup \{h_j\}_{j \in J}$ of $k[G]$. For every $f \in F$, we can then write

$$f \circ m = \sum_{i \in I} a_i \otimes f_i + \sum_{j \in J} b_j \otimes h_j$$

for unique $a_i, b_j \in A$. (Only finitely many a_i and b_j are nonzero.) Specializing this identity along any

$$G \xrightarrow{(x, \mathrm{id})} G \times G$$

gives

$$\lambda_x^{-1}(f) = \sum_{i \in I} a_i(x) f_i + \sum_{j \in J} b_j(x) h_j. \tag{3.1}$$

But $\lambda_x^{-1}(f)$ lies in F by assumption, and any linear combination as in (3.1) is unique because the f_i and h_j form a basis for $k[G]$, so we get $b_j(x) = 0$ for all $j \in J$ and $x \in G$. In other words, $b_j : G \to k$ is the zero map. This means that $f \circ m \in k[G] \otimes_k F$ as claimed. □

Proof of Theorem 3.2. Starting with G, let $f_1, \dots, f_r \in k[G]$ be k-algebra generators. For each, let F_i be a finite-dimensional translation invariant space containing it as in Claim 1. For example, take $F_i = F_{f_i}$. Then $F = \sum_{i=1}^r F_i$ is still finite-dimensional, translation invariant, and contains $f_1, \dots, f_r$. We obtain a homomorphism

$$\begin{aligned} G &\longrightarrow \mathrm{GL}(F) \\ x &\longmapsto \lambda_x \end{aligned} \tag{3.2}$$

and will show that this defines a closed embedding of LAGs.

For concreteness, we first choose a k-basis $h_1, \dots, h_n$ for F. By Claim 2, $m^*(F) \subset k[G] \otimes F$, so there exist unique functions $m_{ji} \in k[G]$ such that

$$m^*(h_i) = \sum_{j=1}^n m_{ji} \otimes h_j. \tag{$\star$}$$

For every $x \in G$, this defines a matrix $\varphi'(x) = (m_{ij}(x))_{i,j} \in M_n(k)$. Evaluating ($\star$) at $(x, y) \in G \times G$, we see

$$h_i(xy) = \sum_{j=1}^n m_{ji}(x) h_j(y).$$

That is, $\varphi'(x)$ expresses the right representation $f \mapsto (\lambda_x^{-1})(f)$ on F as matrix acting from the left in the basis $h_1, \dots, h_n$. So we have the right action rule

$$\varphi'(xy) = \varphi'(y)\varphi'(x).$$

But elements of G are invertible, so φ' factors through $\mathrm{GL}_n \subseteq \mathrm{M}_n(k)$. We can compose it with the inverse and define $\varphi(x) := \varphi'(x)^{-1}$. Since taking the inverse on GL_n is a morphism (see proof of Corollary 3.3 above), and since $(ab)^{-1} = b^{-1}a^{-1}$, this defines a homomorphism of LAGs

$$\varphi : G \longrightarrow \mathrm{GL}_n.$$

Clearly, $\varphi = \lambda_x$ as in (3.2).

Final claim. The $\{m_{ji}\}_{i,j}$ generate $k[G]$ as k-algebra. In particular, φ is a closed immersion.

Proof. We use the identity axiom

$$\begin{array}{ccc} G & \xrightarrow{(\mathrm{id}, e)} & G \times G \\ & \searrow\!\!\searrow & \downarrow m \\ & & G \end{array}$$

which when applied to ($\star$) gives

$$h_i = h_i \circ (\mathrm{id}, e) \circ m = \sum_{j=1}^n m_{ji} h_j(e).$$

Hence the $h_1, \dots, h_n$ lie in the space generated by the $\{m_{ji}\}$, and since the $h_1, \dots, h_n$ spanned F which contains generators for $k[G]$, this means that the m_{ji} generate $k[G]$ as well. □

This concludes the proof of the theorem. □

Remark 3.4. Theorem 3.2 also explains where the term "linear" in LAG comes from:

$$\text{LAGs} \;=\; \text{the closed subgroups of general linear groups.}$$

3.2. **Varieties.** We take a moment to introduce general varieties. This investment will pay off in the future when we study group actions. Today, it will only come up as a more natural setting for constructible sets.

Definition 3.5. A *variety* over k is a pair

$$(X, \mathscr{O}_X)$$

where X is topological space, and where $\mathscr{O}_X$ is the following datum: For every open $U \subseteq X$, a subring

$$\mathscr{O}_X(U) \subseteq \mathrm{Map}(U, k).$$

Here, $\mathrm{Map}(U,k)$ is a ring with $(f+g)(x) = f(x) + g(x)$ and $(fg)(x) = f(x)g(x)$. The function in $\mathscr{O}_X(U)$ are called the *regular* functions on U. We require that $(X, \mathscr{O}_X)$ satisfies the following three axioms:

- Whether or not a function $f : U \to k$ is regular can be checked locally. That is, for all open coverings $U = \bigcup_{i\in I} U_i$,

$$f \text{ is regular} \quad \iff \quad \text{each restrection } f|_{U_i} \text{ is regular.}$$

- Locally, $(X, \mathscr{O}_X)$ is isomorphic to an affine variety. That is, every $x \in X$ has an open neighborhood U such that

$$(U, \mathscr{O}_X|_U) \cong (\mathrm{Sp}(A), \mathscr{O}_{\mathrm{Sp}(A)})$$

for some reduced finite type k-algebra A.

- The topological space X is quasi-compact. That is, there exists a finite open covering of X by opens subsets $\mathrm{Sp}(A_1), \ldots, \mathrm{Sp}(A_r)$.

Remark 3.6. Our notion of variety is equivalent to reduced finite type k-scheme. Traditionally, "variety" also often includes the condition that X is irreducible, but we will work with the more flexible notion.

Exercise. Extend the proof of Proposition 2.21 to show that every closed subset of a variety is a finite union of irreducible subsets. *Hint:* First show that every descending chain of closed subsets of a variety stabilizes.

Definition 3.7. A *morphism of varieties* $\varphi : (Y, \mathscr{O}_Y) \to (X, \mathscr{O}_X)$ is a continuous map $\varphi : Y \to X$ which preserves regular functions. That is, for every two open subsets $U \subseteq X$ and $V \subseteq Y$ with $\varphi(V) \subseteq U$, and for every regular function $f : U \to k$, the composition

$$V \xrightarrow{\varphi} U \xrightarrow{f} k$$

is again regular.

Trivial example: An open or closed subsets of a variety again has a natural variety structure. These are the *open* resp. *closed* subvarieties.

Definition 3.8. A subset $W \subseteq X$ of a topological space is called *locally closed* if it can be written as the intersection of an open and a closed subsets. (It is closed in an open subset, hence the terminology.)

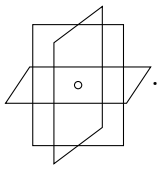
If $(X, \mathscr{O}_X)$ is a variety, then every locally closed subset $W = U \cap Z$ has a natural variety structure $(W, \mathscr{O}_W)$, and W is said to be a *subvariety*. This simply combines the open and the closed case:

$$U \cap Z \underset{\text{closed subvar.}}{\subseteq} U \underset{\text{open subvar.}}{\subseteq} X.$$

Example 3.9. $k^2 \smallsetminus \{0\} = D(x) \cup D(y)$ is an open subvariety of $k^2 = \mathrm{Sp}(k[x,y])$ covered by two affine opens. But it is not itself affine.

Example 3.10. The union of the three coordinate planes minus the origin is a (non-affine) subvariety of k^3:

$$V(xyz) \smallsetminus \{(0,0,0)\} \subset k^3$$



Question. Why care for varieties when we are mostly interested in LAGs which are affine?

Answer. Quotients of LAGs are often not affine anymore.

Example 3.11. Consider $G = \mathrm{GL}_2$, $X = k^2 \smallsetminus \{(0,0)\}$, and the action

$$G \times X \xrightarrow{\mu} X, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} := \begin{pmatrix} ax+by \\ cx+dy \end{pmatrix}.$$

Note that μ is a morphism of varieties because it comes by restriction from a polynomial map $G \times k^2 \to k^2$.

Fix the vector $x := (1\ 0)^t \in X$. We observe that

- X is single orbit, i.e. $G \cdot x = X$ (as set), and that
- the stabilizer

$$G_x = \{g \in G \mid gx = x\} = \underbrace{\mu^{-1}(x) \cap G \times \{x\}}_{\text{both closed}}$$

is closed subgroup of G, hence again a LAG. In this case:

$$G_x = \left\{ \begin{pmatrix} 1 & b \\ & d \end{pmatrix} \right\} \subseteq \mathrm{GL}_2.$$

So we see that

$$\underbrace{G/G_x \xrightarrow{\sim} X}_{\text{will be formalized later}} \qquad \text{is a non-affine variety!}$$

3.3. **Constructible subsets.** Reference: [1, §4.3, 4.4].

Question: Let $\varphi : Y \to X$ be a morphism of varieties. Which properties does $\varphi(Y)$ have?

Theorem 3.12. *The possible images $\varphi(Y)$ for varying Y, φ are precisely the* constructible subsets *of X.*

Definition 3.13. A subset $W \subseteq X$ is called *constructible* if it can be written as a finite union of locally closed subsets.

The constructible subsets are precisely those which can be obtained from open subsets and finitely many operations of the form taking union, intersection, or complement.

Example 3.14. A constructible set which is not locally closed would be

$$\big(V(xyz) \cap D(x)\big) \cup \{(0,0,0)\} \subset k^3.$$

Proof of Theorem 3.12. Clearly, if $W = W_1 \cup \cdots \cup W_r$ is finite union of locally closed subsets W_i, then each W_i is variety and

$$W = \mathrm{Im}\left(\coprod_{i=1}^{r} W_i \longrightarrow X \right)$$

is trivially the image of a morphism. The meaningful direction is the converse. So let $\varphi : Y \to X$ be a morphism, our goal being to show that $\varphi(Y)$ is constructible.

Step 1: Reduction to affine case We cover X by finitely many affine varieties, $X = U_1 \cup \ldots \cup U_r$, $U_i \cong \mathrm{Sp}(A_i)$. Then

$$\varphi(Y) = \bigcup_{i=1}^{r} \varphi(\varphi^{-1}(U_i))$$

and it is enough to show that every set on the right hand side is constructible. This allows to assume X affine from now on.

Similarly, if $Y = V_1 \cup \ldots \cup V_r$ is a finite covering by affines, then

$$\varphi(Y) = \bigcup_{i=1}^{r} \varphi(V_i),$$

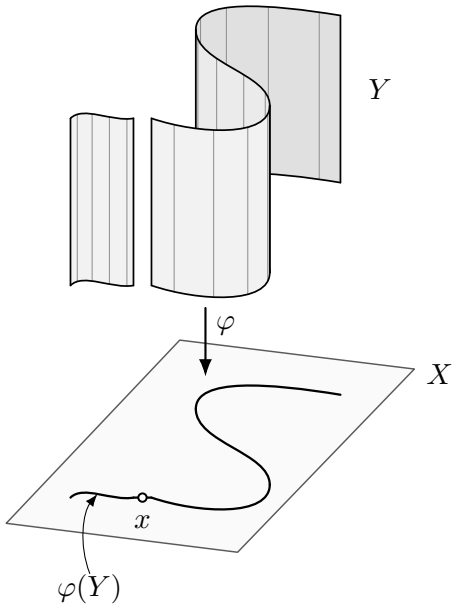
so we may also assume that Y is affine.

Step 2: Reduction to integral domains Write Y as a union of its irreducible components $Y_1, \ldots, Y_r$. Then

$$\varphi(Y) = \bigcup_{i=1}^{r} \varphi(Y_i),$$

so it suffices to consider the case when Y is irreducible. Then the image $\varphi(Y) \subseteq X$ is irreducible as well. It follows that the closure $\overline{\varphi(Y)}$ is irreducible as well. This closure is a closed subvariety of X, and it suffices to show that $\varphi(Y) \subseteq \overline{\varphi(Y)}$ is constructible.

Here is a sketch of the situation,



with $\overline{\varphi(Y)} = \varphi(Y) \cup \{x\}$

Replacing X by $\overline{\varphi(Y)}$, we have reduced to irreducible $Y = \mathrm{Sp}(B)$, $X = \mathrm{Sp}(A)$, and a morphism φ with dense image $\varphi(Y) \subseteq X$. Then A and B are integral domains by (2.1), and the pullback map $\varphi^* : A \to B$ is injective. Indeed, $\varphi^*(f) = 0$ means that φ factors through the vanishing set $V(f)$ which would be a *proper* closed subset if $f \neq 0$.

The main argument. Everything so far was formal and immediate, but now the actual work starts. Recall that $\varphi^* : A \hookrightarrow B$ is an injective map of integral domain finite type k-algebras.

Consider the field extension $\mathrm{Quot}(A) \subseteq \mathrm{Quot}(B)$. It is generated by B and has finite transcendence degree, say $r \geq 0$. Choose a transcendence basis $b_1, \ldots, b_r \in B$. Let $A[b_1, \ldots, b_r] \subseteq B$ be the A-subalgebra generated by $b_1, \ldots, b_r$. Since the b_i are algebraically independent, each of its elements is a unique polyonomial in the $b_1, \ldots, b_r$ with coefficients in A. In this way, we have factored φ as $p_X \circ \psi$:

$$\begin{array}{ccccc} Y & \xrightarrow[\psi]{((b_1,\ldots,b_r),\varphi)} & k^r \times X & \xrightarrow{\quad p_X \quad} & X \\ B & \xleftarrow{\qquad\qquad} & A[b_1,\ldots,b_r] & \xleftarrow{\qquad} & A \end{array}$$

Step 3. We claim that $\psi(Y)$ contains an open subset of $k^r \times X$.

Proof. Since B is a finitely generated k-algebra, it is also a finitely generated $A[b_1, \ldots, b_r]$-algebra. Let $\beta_1, \ldots, \beta_m$ be a set of generators. Since $\mathrm{Quot}(B)/\mathrm{Quot}(A)(b_1, \ldots, b_r)$ is finite, each β_i is the zero of some monic polynomial

$$T^{d_i} + c_{i,d_i-1}T^{d_i-1} + \cdots + c_{i,0},$$

with coefficients of the form $c_{i,j} = \frac{p_{i,j}}{q_{i,j}}$ for some $p_{i,j}, q_{i,j} \in A[b_1, \ldots, b_r]$. Let

$$q := \prod_{i,j} q_{i,j}$$

be the product of all denominators. Then

- $B[q^{-1}]$ is still generated by $\beta_1, \ldots, \beta_m$ as $A[b_1, \ldots, b_r, q^{-1}]$-algebra.
- Each β_i is integral over $A[b_1, \ldots, b_r, q^{-1}]$.

So $A[b_1, \ldots, b_r, q^{-1}] \subseteq B[q^{-1}]$ is a finite extension of integral domains, and hence

$$\mathrm{Sp}(B[q^{-1}]) \longrightarrow\!\!\!\!\!\rightarrow \mathrm{Sp}(A[b_1, \ldots, b_r, q^{-1}])$$

surjective by going up and going down. We see that $D(q) \subseteq k^r \times X$ lies in the image. □

Step 4. Consider an irreducible $X = \mathrm{Sp}(A)$ as before, the projection $p_X : k^r \times X \to X$, and a non-empty open subset $U \subseteq k^r \times X$. We claim that $p_X(U)$ contains an open subset of X.

Proof. Let $0 \neq q \in A[b_1, \ldots, b_r]$ be a polynomial such that $D(q) \subseteq U$. Express it in standard form

$$q = \sum_{I \in \mathbb{Z}^r_{\geq 0}} a_I b^I.$$

Then $p_X(D(q)) = \bigcup_I D(a_I)$ because precisely for the points x of this union is $q(-, x) \in k[b_1, \ldots, b_r]$ a non-zero polynomial and equivalently

$$D(q(-, x)) = k^r \times \{x\} \cap D(q)$$

non-empty. □

Step 5: End by induction. Combining Steps 3 and 4, we have shown that $\varphi(Y) \subseteq X$ contains an open subset. Since X is irreducible, $X \setminus U$ is of strictly smaller dimension than X. We end by induction. □

3.4. **Images are again LAGs.** Reference: [1, §7.4]

After these preparations, we can can wrap up the argument that images of homomorphisms are automatically closed.

Proposition 3.15. *Let $\varphi : G \to H$ be a homomorphism of LAGs. Then $\varphi(G) \subseteq H$ is a closed subgroup and, in particular, again naturally an affine variety.*

Proof. By Theorem 3.12, the image $\varphi(G) \subset H$ is constructible and a subgroup. We need a few lemmas.

Lemma 3.16. *Let $K \subset H$ be a set-theoretic subgroup of a LAG H. Then the closure $\overline{K} \subseteq H$ is a subgroup as well.*

Proof. By Corollary 3.3, the inverse map $i : H \longrightarrow H$ is a variety automorphism. In particular, it is a homeomorphism, so

$$\overline{K}^{-1} = \overline{K^{-1}} = \overline{K}.$$

This shows that $\overline{K}$ is stable under taking inverse. For $x \in K$, $x \cdot \overline{K} = \overline{xK} = \overline{K}$ since left multiplication by x is homeomorphism as well. Hence $K\overline{K} \subseteq \overline{K}$.

As a special case, this says that for $x \in \overline{K}$ we have $Kx \subseteq \overline{K}$. Since right multiplication by x is also a homeomorphism, we get that $\overline{Kx} = \overline{K}x \subseteq \overline{K}$. So after all $\overline{K} \cdot \overline{K} \subseteq \overline{K}$, showing that $\overline{K}$ is a group as claimed. □

Lemma 3.17. *Let X be a variety and $W \subseteq X$ a constructible subset. Then W contains a dense open subset of its closure $\overline{W}$.*

Proof. Left as exercise. □

Lemma 3.18. *Let $U, V \subseteq K$ be open subsets of LAG K with U being dense. Then $U \cdot V := \mathrm{Im}(U \times V \xrightarrow{m} K) = K$.*

Proof. For $x \in K$, by Corollary 3.3, the translated inverse $x \cdot i(V)$ is again an open subset. Since U is dense, the intersection $U \cap x \cdot i(V)$ is non-empty. This means there exist $u \in U$ and $v \in V$ such that $u = x \cdot v^{-1}$ as desired. □

Combining the various statements, we first obtain from Lemma 3.16 that $\overline{\varphi(G)} \subseteq H$ is closed. By Lemma 3.17, there exists an open dense subset $U \subseteq \overline{\varphi(G)}$ which is already contained in $\varphi(G)$. By Lemma 3.18, we have $U \cdot U = \overline{\varphi(G)}$. Since $\varphi(G)$ is a group to begin with, we get $\overline{\varphi(G)} \subseteq \varphi(G) \cdot \varphi(G) = \varphi(G)$ as desired. □

4. Actions and orbits (10/10/2026)

General problem. Let G be a LAG and $H \subset G$ a closed subgroup. How to endow the set of orbits G/H with a natural variety structure?

Definition 4.1. An *action of G* on a variety X is defined as a morphism

$$\mu : G \times X \longrightarrow X$$

that is also a set-theoretic action.

Definition 4.2. A *categorical quotient* for G acting on X is defined as a *universal* G-invariant morphism $\pi : X \to Q$. That is, for every G-invariant morphism $\psi : X \to Y$, there exists a unique morphism $\overline{\psi} : Q \to Y$ factoring ψ as

$$\begin{array}{ccc} X & \xrightarrow{\pi} & Q \\ & \searrow^{\psi} & \downarrow{\scriptstyle \overline{\psi}} \\ & & Y. \end{array}$$

In the above, *G-invariant* means that $\pi(gx) = \pi(x)$ for all $g \in G, x \in X$.

Remark 4.3. A categorical quotient (Q, π) is unique up to unique isomorphism because it is characterized by its universal property.

Definition 4.4. A categorical quotient $\pi : X \to Q$ is called a *geometric quotient* if π also provides a set-theoretic bijection

$$G\backslash X \xrightarrow{\sim} Q.$$

Exercise 4.5. The mutiplicative group acts on the affine line $\mathbb{A}^1$ by scaling, i.e.

$$\mathbb{G}_m \times \mathbb{A}^1 \longrightarrow \mathbb{A}^1, \quad \lambda \cdot x = \lambda x.$$

There are two orbits,

$$k^\times \quad \text{and} \quad \{0\}.$$

Show that the categorical quotient exists, but is equal to a point $\mathrm{Sp}(k) = \{\mathrm{pt}\}$. It is hence not a geometric quotient.

Precise problem. For a subgroup $H \subset G$, prove that the categorical quotient of G by H exists and is a geometric quotient. By Remark 4.3, this defines an unambiguous variety structure on G/H.

Special case. In the previous lecture, we showed that if $H = \ker(\varphi)$ is the kernel of a homomorphism of LAGs $\varphi : G \longrightarrow K$, then the image

$$K \overset{\text{closed}}{\supseteq} \varphi(G) = G/H$$

is closed and hence has a natural variety structure. Even then it is not obvious to see that $G \to \varphi(G)$ is a categorical quotient (for example, independent of φ). Moreover, this construction only applies if H is a kernel, meaning it at least has to be a normal subgroup.

We proceed in three steps:

(1) Given $H \subset G$, we will construct a variety X with G-action such that H is the stabilizer of some $x \in X$.

(2) The set-quotient G/H can then be identified with the orbit $G \cdot x$. We prove that that $G \cdot x$ is locally closed in X which gives it a variety structure.

(3) We use two more advanced algebraic geometry results to show that this makes $G \cdot x$ into a categorical quotient.

4.1. **Orbits.** Reference: [1, §8.1-§8.3]

Throughout, let G be a LAG acting on a variety X. Then, at the level of sets,

$$X = \coprod_{G\cdot x \,\in\, G\backslash X} G \cdot x$$

is the disjoint union of its orbits. Each $x \in X$ has a stabilizer subgroup $G_x \subseteq G$, and there is a bijection

$$G/G_x \xrightarrow{\sim} G \cdot x.$$

Lemma 4.6. *Each stablizer $G_x \subseteq G$ is a closed subgroup.*

Proof. Points of varieties are always closed. (Being closed can be checked locally for an open covering, and points of affine varieties are closed.) Consider the orbit morphism

$$\mu_x : G \longrightarrow X, \qquad \mu_x(g) = g \cdot x.$$

Then G_x can be described as the inverse image

$$G_x = \mu_x^{-1}(\{x\}) \subseteq G$$

and is hence closed. □

Proposition 4.7. *(a) Each $G \cdot x$ is locally closed in X and smooth with its subvariety structure.*

(b) Each orbit is equidimensional, meaning all its irreducible components have the same dimension. This dimension is given by

$$\dim(G \cdot x) = \dim(G) - \dim(G_x).$$

(c) The closure $\overline{G \cdot x}$ of an orbit is G-stable. The boundary $\overline{G \cdot x} \setminus G \cdot x$ is a union of G-orbits of strictly smaller dimension than $G \cdot x$.

(d) Each G-orbit is the disjoint union of its (finitely many) G°-orbits.

Proof. (a) By Theorem 3.12, $G \cdot x = \mu_x(G)$ is a constructible subset of X. Hence there exists an open subset $U \subseteq \overline{G \cdot x}$ contained in $G \cdot x$ (Lemma 3.17). That is, there is an open subset $\widetilde{U} \subseteq X$ such that the intersection

$$U = \widetilde{U} \cap \overline{G \cdot x}$$

is contained in $G \cdot x$. We define

$$\widetilde{V} := \bigcup_{g \in G} g \cdot \widetilde{U}, \quad \text{and} \quad V = \widetilde{V} \cap \overline{G \cdot x}.$$

Then $\widetilde{V}$ is open, so V is locally closed. But we also have

$$V = \bigcup_{g \in G} g \cdot U = G \cdot x.$$

Hence $G \cdot x$ is locally closed in X and, in particular, a subvariety. For the smoothness, we would like to apply the following proposition.

Proposition 4.8. *Let $\varphi : Z \to Y$ be a dominant morphism of varieties over a field of characteristic 0. Then there exists a point $z \in Z$ in which φ is smooth.*

Here, *dominant* just means that the image dense: $\overline{\varphi(Z)} = Y$. This is obviously true for $G \mapsto G \cdot x$ which is even surjective. Before continuing, we recall the definition of smoothness:

Definition (Smoothness). A morphism of varieties $\varphi : Z \to Y$ is smooth in $z \in Z$ if for some choice of affine local charts

$$z \in \operatorname{Sp}(B) \underset{\text{open}}{\subseteq} Z \quad \text{and} \quad \varphi(z) \in \operatorname{Sp}(A) \underset{\text{open}}{\subseteq} Y \tag{4.1}$$

such that $\varphi(\operatorname{Sp}(B)) \subseteq \operatorname{Sp}(A)$, and some choice of presentation

$$B = A[T_1, \ldots, T_n]/(f_1, \ldots, f_m), \tag{4.2}$$

the rank of the Jacobi matrix in z,

$$J_z := \left(\frac{\partial f_i}{\partial T_j} \right)_{i,j} (z) \in M_n(k),$$

equals the codimension of the fiber $\varphi^{-1}(\varphi(z))$ in z:

$$\begin{aligned}\operatorname{codim}_z\big(\varphi^{-1}(\varphi(z)) \overset{T_1,\dots,T_n}{\subset} \mathbb{A}^n\big) &:= n - \dim_z(\varphi^{-1}(\varphi(z))) \\ &:= n - \text{Krull. dim}\big(B/\mathfrak{m}_{\varphi(z)}B\big)_{\mathfrak{m}_z}.\end{aligned}$$

There are several equivalent definitions of smoothness. The above is called the *Jacobi criterion*. A beautiful fact is that it does not depend on the choices made in (4.1) and (4.2). That is, if for one such set of choices the Jacobi criterion is satisfied, then it is satisfied for all such choices.

Smoothness is a *relative* notion in that it is a property of morphisms. If we look at the *absolute* case $Y = \operatorname{Sp}(k)$, then we have the following fact.

Fact. A variety Z over an algebraically closed field k is smooth if and only if all its local rings are *regular*. That is, for every $z \in Z$ and every choice of affine open neighborhood $z \in \operatorname{Sp}(A) \subseteq Z$, the local ring $A_{\mathfrak{m}_z}$ is a regular local ring.

Back to the proof of Proposition 4.7.

(a) We have shown that any orbit $G \cdot x \subseteq X$ is locally closed and hence a subvariety. By Proposition 4.8, there exists a smooth point $y \in G \cdot x$. Since G acts on $G \cdot x$ transitively, every point of $G \cdot x$ is smooth. □ (a)

(b) Proposition 4.8 also gives that the orbit map

$$\mu_x : G \longrightarrow G \cdot x, \qquad g \longmapsto gx, \tag{4.3}$$

is smooth. Namely, there exists a smooth point $g \in G$ for μ_x. Left translation by an element $h \in G$ defines a commutative square

$$\begin{array}{ccc} G & \xrightarrow[\sim]{h\,\cdot\,-} & G \\ \downarrow & & \downarrow \\ G \cdot x & \xrightarrow[\sim]{h\,\cdot\,-} & G \cdot x. \end{array}$$

The horizontal arrows being isomorphisms, this shows that μ_x is also smooth in hg. Since h was arbitrary, we obtain smoothness of μ_x in every point.

In general, if a morphism $\varphi : Z \to Y$ is smooth in a point z then

$$\dim_z(Z) = \dim_z(\varphi^{-1}(\varphi(z))) + \dim_{\varphi(z)}(Y).$$

In our setting, the fiber over $g \cdot x$ is the translate gG_x, which is isomorphic to G_x, so we get

$$\dim_g(G) = \dim_g(\underbrace{gG_x}_{\mu_x^{-1}(gx)}) + \dim_{gx}(G \cdot x).$$

This directly yields (b). □ (b)

(c) For each $g \in G$, left translation $X \xrightarrow{g} X$ is a homeomorphism and hence commutes with taking closure. This means that

$$g \cdot \overline{G \cdot x} = \overline{gG \cdot x} = \overline{G \cdot x}$$

which says that $\overline{G \cdot x}$ is G-stable. The claim about the dimension of the boundary is automatic. In general, for a constructible subset W of a variety X,

$$\dim(\overline{W}) = \dim(W) \qquad \text{and} \qquad \dim(\overline{W} \setminus W) < \dim(W).$$

Here, we define $\dim(W)$ as $\max \dim(W_i)$ if $W = \bigcup_i W_i$ with W_i locally closed. □ (c)

(d) Viewed as sets, we can decompose a G-orbit into its G°-orbits:

$$G \cdot x = \bigcup_{y \in G/G^\circ} G^\circ \cdot yx$$

The claim is that each $G^\circ \cdot yx \subseteq G \cdot x$ is open and closed for the subspace topology.

Closedness. If $z \in \overline{G^\circ \cdot yx} \setminus G^\circ \cdot yx$, then $\dim(G^\circ \cdot z) < \dim(G^\circ \cdot yx)$ by (c) applied to G°. But

$$\dim(G \cdot z) = \dim(G^\circ \cdot z) \quad \text{and} \quad \dim(G \cdot x) = \dim(G^\circ \cdot yx)$$

because the G-orbits are finite unions of homeomorphic translates of the G°-orbits. So we get $z \in \overline{G \cdot x} \setminus G \cdot x$ by (c) for G. This means that $G^\circ \cdot yx$ is closed in $G \cdot x$.

Openness. This follows from closedness by taking complements:

$$G^\circ \cdot yx = G \cdot x \setminus \bigcup_{G^\circ \cdot y'x \neq G^\circ \cdot yx} G^\circ \cdot y'x.$$

□

Corollary 4.9. *For every action $G \times X \to X$, there exists at least one closed orbit.*

Proof. Begin with any orbit $G \cdot x$. If it is not closed, replace x by some $x' \in \overline{G \cdot x} \setminus G \cdot x$. Then $\dim(G \cdot x') < \dim(G \cdot x)$ by Proposition 4.7 (c). Iterating this procedure will at some point result in a closed orbit. □

4.2. **Natural examples of group actions.** In the second half of the lecture, we want to discuss some examples.

Example 4.10. Consider the action of $\mathbb{G}_m$ on $\mathbb{A}^n$ given by

$$\lambda \cdot (x_1, \ldots, x_n) = (\lambda x_1, \ldots, \lambda x_n).$$

The orbits are $\{0\}$ which is closed and the sets $\ell \setminus \{0\}$ where $\ell = k \cdot v \subseteq k^n$ is a line. If $n \geq 2$, the orbits of the second kind are neither open nor closed.

Example 4.11. Consider $\mathbb{G}_m$ action on $\mathbb{A}^2$, this time by the formula

$$\lambda \cdot (x, y) = (\lambda x, \lambda^{-1} y).$$

There are three types of orbits:

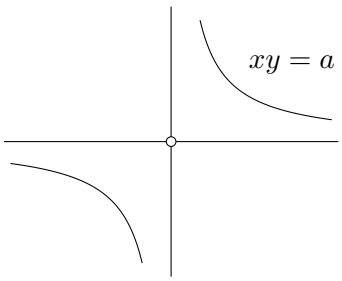
(1) The origin $\{(0,0)\}$, which is closed;

(2) The 1-dimensional hyperbolas $\{xy = a \mid a \in k^\times\}$, which are also closed since they are defined by the vanishing of $xy - a$;

(3) The 1-dimensional orbits given by the open parts of the coordinate axes:

$$\{x \neq 0, y = 0\} \quad \text{and} \quad \{x = 0, y \neq 0\}.$$

These are only locally closed.



Example 4.12. Let V be a finite-dimensional k-vector space, viewed as variety, and let $G = \mathrm{GL}(V)$, viewed as algebraic group. The standard action $\mathrm{GL}(V) \curvearrowright V$ has two orbits: $\{0\}$, which is closed, and $V \setminus \{0\}$, which is open.

Assume now that $\dim(V) \geq 2$ and let $\mathrm{GL}(V) \curvearrowright V^2$ diagonally. That is, we define

$$g \cdot (v, w) := (gv, gw).$$

There are three types of orbits:

(1) The open $2n$-dimensional orbit

$$\{(v, w) \mid v, w \text{ linearly independent}\}.$$

(2) For every slope $a \in k \cup \{\infty\}$, an n-dimensional locally closed orbit given by the condition that v and w are related by factor a:

$$\ell_a^\times = \begin{cases} \{(v, av)\} & v \neq 0 \text{ if } a \in k \\ \{(0, w)\} & w \neq 0 \text{ if } a = \infty. \end{cases}$$

(3) The closed orbit $\{(0, 0)\}$.

In particular, (1) is an example of an orbit $G \cdot x$ such that $\overline{G \cdot x} \setminus G \cdot x$ consists of infinitely many smaller G-orbits.

We consider the special case $n = 2$. An further observation in this situation is that

$$v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}, \quad w := \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$$

are linearly independent if and only if $\det(v, w) = v_1 w_2 - v_2 w_1 \neq 0$. So we have a morphism $\det : V^2 \to \mathbb{A}^1$ which is GL_2-equivariant in the sense

$$\det(gv, gw) = \det(g) \det(v, w).$$

The preimage $\det^{-1}(k^\times)$ is an open orbit. The preimage $\det^{-1}(0)$ is the quadric

$$v_1 w_2 - v_2 w_1 = 0.$$

which is the union of all ℓ_a, for $a \in k \cup \{\infty\}$. If $a \neq b$, then ℓ_a and ℓ_b intersect in $\{0\}$.

Example 4.13 (Standard representation of SO_n)**.** We consider the standard representation of the special orthogonal group. That is, let $H \in \mathrm{Sym}_n(k)$ be a non-degenerate symmetric matrix. It defines a symmetric bilinear pairing $(v, w) := v^t H w$ on k^n. The associated quadratic form is

$$Q(v) := \frac{1}{2} v^t H v$$

where the factor $1/2$ is a standard normalizing constant. (We only put it by convention, it has no mathematial effect.) The special orthogonal group $G := \mathrm{SO}(H)$ acts on k^n which is called the *standard representation*. Clearly, the morphism

$$Q : \mathbb{A}^n \longrightarrow k$$

defined by Q is G-invariant. In fact, we claim that if $v, w \in k^n \setminus \{0\}$, then

$$G \cdot v = G \cdot w \quad \iff \quad Q(v) = Q(w).$$

That is, the G-orbits of k^n are precisely

$$k^n = \{0\} \sqcup \{v \neq 0 \mid Q(v) = 0\} \sqcup \bigsqcup_{a \in k^\times} \{v \mid Q(v) = a\}.$$

One direction here is clear: If $w = gv$, then $Q(v) = Q(w)$. Conversely, assume v, w nonzero with $Q(v) = Q(w) =: q$.

• First assume we are in the so-called *anisotropic* case, meaning $q \neq 0$. Then by taking orthogonal complement the vector space k^n decomposes as

$$k^n = \langle v \rangle \oplus \langle v \rangle^\perp = \langle w \rangle \oplus \langle w \rangle^\perp.$$

Both $\langle v \rangle^\perp$ are $\langle w \rangle^\perp$ are non-degenerate quadratic spaces of the same dimension over the algebraically closed field k. Using the Gram–Schmidt method to construct an orthonormal basis for both, we see that they are isomorphic:

$$\langle v \rangle^perp,\ \langle w \rangle^\perp \ \cong (k^{n-1}, \mathbb{1}_{n-1}).$$

So we can choose an isometry

$$\varphi : \langle v \rangle^\perp \xrightarrow{\sim} \langle w \rangle^\perp.$$

Then $g : k^n \longrightarrow k^n$, defined by

$$g(v) = w, \qquad g|_{\langle v \rangle^\perp} = \varphi$$

lies in $O(k^n, H)$. Namely, for $\lambda \in k$ and $v^\perp \in \langle v \rangle^\perp$

$$\begin{aligned}(g(\lambda v + v^\perp), g(\lambda v + v^\perp)) &= \lambda^2(gv, gv) + 2\lambda(gv, gv^\perp) + (gv^\perp, gv^\perp) \\ &= \lambda^2(w, w) + 2\lambda(w, \varphi(v^\perp)) + (\varphi(v^\perp), \varphi(v^\perp)) \\ &= \lambda^2(v, v) + 0 + (v^\perp, v^\perp) \\ &= (\lambda v + v^\perp, \lambda v + v^\perp).\end{aligned}$$

It might be that $\det(g) = -1$. In this situation, we can change its sign on a vector of an orthonormal basis of $\langle v \rangle^\perp$ to ensure that $g \in G$. We conclude that $G \cdot v = G \cdot w$.

• It is left ot consider the *isotropic* case $q = 0$. The argument will be similar, but we cannot take the orthogonal complement of $\langle v \rangle$ and $\langle w \rangle$ to decompose V. Instead, we argue that since $(\ ,\ )$ is non-degenerate, there exists an element $e' \in k^n$ such that

$$(v, e') \neq 0.$$

Recaling e' if necessary, we can assume

$$(v, e') = 1.$$

Then for every $\lambda \in k$,

$$(e' + \lambda v, e' + \lambda v) = (e', e') + 2\lambda.$$

So

$$e := e' - \frac{(e', e')}{2} v$$

is isotropic, $(e, e) = 0$, and still satisfies $(v, e) = 1$. Hence

$$\langle v, e \rangle \cong \left(k^2, \begin{pmatrix} & 1 \\ 1 & \end{pmatrix} \right)$$

is a *hyperbolic plane*. In particular, it is a non-degenerate subspace, so we may take its orthogonal complement to decompose k^n. Performing the same construction for w, we find two decompositions

$$k^n = \langle v, e_v \rangle \oplus \langle v, e_v \rangle^\perp = \langle w, e_w \rangle \oplus \langle w, e_w \rangle^\perp.$$

We can now construct $g \in G$ with $gv = w$ as before. □ Claim

Remark 4.14. The morphism $Q : k^n \to k$ is a categorical quotient $\mathrm{SO}(H)\backslash k^n$. Translated to commutative algebra, this means that if $f \in k[k^n]$ is a $\mathrm{SO}(H)$-invaraint polynomial, then there exists a polynomial $r \in k[T]$ such that

$$f(T_1, \ldots, T_n) = r(Q(T_1, \ldots, T_n)).$$

The map is not quite a geometric quotient though:

$$Q^{-1}(q) = \begin{cases} \text{single closed orbit} & q \neq 0 \\ \{0\} \sqcup \{ \text{ isotropic vectors}\} & q = 0. \end{cases}$$

Example 4.15 (Conjugation on $(n \times n)$-matrice)**.** Consider the action of GL_n on $\mathrm{M}_n(k)$ by conjugation,

$$g \cdot x := gxg^{-1}.$$

We first study the case $n = 2$. There exists a G-invariant morphism

$$\mathrm{M}_2(k) \longrightarrow k^2, \qquad x \longmapsto \big(\mathrm{tr}(x), \det(x)\big).$$

We know that the characteristic polynomial of matrix x,

$$\mathrm{char}(x, T) = T^2 - \mathrm{tr}(x)T + \det(x)$$

is separable if and only if the discriminant

$$\Delta(x) := \mathrm{tr}(x)^2 - 4\det(x) \neq 0.$$

The discriminant Δ is a morphism $k^2 \to k$,

$$\Delta(t, d) := \Delta(T^2 + tT + d) = t^2 - 4d.$$

For (t, d) with $\Delta(t, d) \neq 0$, we have a closed 2-dimensional orbit

$$X_{t,d} := \{x \in \mathrm{M}_2(k) \mid \mathrm{tr}(x) = -t, \ \det(x) = d\}.$$

Here, we are using our knowledge from linear algebra: Two matrices x, y with separable characteristic polynomial are conjugate if and only if their characteristic polynomials are equal.

We now consider the non-separable case $4d = t^2$. Then $X_{t,d}$ breaks up according to the possible Jordan normal forms:

$$X_{t,d} = \underbrace{\mathrm{GL}_2 \cdot \begin{pmatrix} t/2 & 1 \\ & t/2 \end{pmatrix}}_{\text{2-dimensional orbit}} \sqcup \underbrace{\left\{ \begin{pmatrix} t/2 & \\ & t/2 \end{pmatrix} \right\}}_{\substack{\text{closed orbit,} \\ \text{0-dimensional}}}.$$

This picture generalizes beautifully to general n. First, trace and determinant are generalized by the morphism

$$\mathrm{char} : \mathrm{M}_n(k) \longrightarrow k^n, \qquad x \longmapsto (a_{n-1}, \ldots, a_0) \tag{4.4}$$

taking x to the coefficients of its characteristic polynomial

$$\mathrm{char}(x, T) = T^n + a_{n-1}T^{n-1} + \cdots + a_0.$$

Observe that each a_i is a polynomial in the n^2 entries of x, so this is indeed a morphism of algebraic varieties. We claim there exists a discriminant polynomial

$$\Delta : k^n \longrightarrow k \tag{4.5}$$

such that

$$\Delta(a_{n-1}, \ldots, a_0) \neq 0$$

if and only if the polynomial

$$T^n + a_{n-1}T^{n-1} + \cdots + a_0.$$

is separable. This statement is not immediate. One way to construct Δ is the following. First, we consider the morphism

$$X := k^n \longrightarrow k^n, \quad (\lambda_1, \ldots, \lambda_n) \longmapsto \prod_{i=1}^{n}(T - \lambda_i).$$

The right hand side here is meant as "take the vector of coefficients of this polynomial". Each of these coefficients is a polynomial in the λ_i, and hence this is indeed a morphism $X \to \mathbb{A}^n$.

The reason for introducing this "covering" of $\mathbb{A}^n$ is that it is trivial to encode separability in terms of the zeroes of a polynomial. Consider

$$\delta : X \longrightarrow k, \quad (\lambda_1, \ldots, \lambda_n) \longmapsto \prod_{i \neq j}(\lambda_i - \lambda_j). \tag{4.6}$$

Then

$$\delta(\lambda_1, \ldots, \lambda_n) \neq 0 \quad \Longleftrightarrow \quad \prod_{i=1}^{n}(T - \lambda_i) \text{ separable.}$$

The group S_n acts on X by permuting the λ_i.

Theorem 4.16 (Theorem of elementary symmetric functions)**.** *The S_n-invariant morphism*

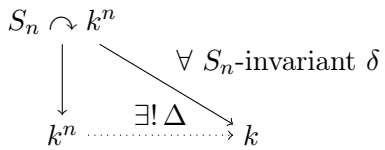
$$X \longrightarrow \mathbb{A}^n$$

$$(\lambda_1, \ldots, \lambda_n) \longmapsto \textit{coeff. of } \prod_{i=1}^{n}(T - \lambda_i)$$

is a categorical quotient $S_n \backslash X$. That is, for every S_n-invariant polynomial $\delta \in k[\lambda_1, \ldots, \lambda_n]$, there exists a polynomial $\Delta \in k[a_{n-1}, \ldots, a_0]$ such that

$$\delta(\lambda_1, \ldots, \lambda_n) = \Delta\big(\textit{coeff. of } \prod_{i=1}^{n}(T - \lambda_i)\big).$$

That is, we have the diagram situation



Applying the theorem to the δ from (4.6), we find the desired discriminant morphism $\Delta : k^n \to k$ such that

$$T^n + a_{n-1}T^{n-1} + \ldots + a_0 \text{ separable } \Longleftrightarrow \ \Delta(a_{n-1}, \ldots, a_0) \neq 0.$$

Coming back to (4.4), we find that for all polynomials in the principal open $D(\Delta) \subset k^n$,

$$\begin{aligned} X_P &:= \operatorname{char}^{-1}(P) \\ &= \{x \in \mathrm{M}_n(k) \mid \operatorname{char}(x; T) = P(T)\} \end{aligned}$$

is a single closed conjugation orbit. In general, X_P is a finite union of orbits described by the possible Jordan normal forms for characteristic polynomial P.

5. Two case studies

We analyze two important examples in more detail. These examples motivate a lot of the general theory of reductive groups which we'll discuss later in the course. For now, their role is to add more color to the abstract definitions we've seen so far.

5.1. **Case study 1: The nilpotent cone in $\mathrm{M}_n(k)$.** The so-called *nilpotent cone* in $\mathrm{M}_n(k)$ is the variety of nilpotent matrices

$$X := \{x \in \mathrm{M}_n(k) \mid x^n = 0\}.$$

The group GL_n acts on X by conjugation. Our goal is to completely describe the orbits and their closure relations. This extends Example 4.15 in the most extreme case where the characteristic polynomial is T^n.

Orbits. Every $x \in \mathrm{M}_n(k)$ has a Jordan normal form, and two matrices are conjugate if and only if their Jordan normal forms agree up to a permutation of blocks. A nilpotent matrix has 0 as its only eigenvalue, so its Jordan normal form is determined by the sizes of its Jordan blocks. This gives a bijection

$$\mathrm{GL}_n\backslash X \overset{1:1}{\longleftrightarrow} \{\text{partitions } n = n_1 + \cdots + n_k \text{ with } n_1 \geq \cdots \geq n_k > 0\}$$

$$\mathrm{GL}_n \cdot \begin{pmatrix} J_{n_1} & & \\ & \ddots & \\ & & J_{n_k} \end{pmatrix} \longleftarrow\!\shortmid\ n = n_1 + \cdots + n_k.$$

Here,

$$J_m = \begin{pmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{pmatrix} \in \mathrm{M}_m(k)$$

denotes the nilpotent Jordan block of size m. We write $X_{n_1+\cdots+n_k} \subseteq X$ for the orbit corresponding to the partition $n = n_1 + \cdots + n_k$.

Dimensions of orbits. Recall from Proposition 4.7 (b) that

$$\dim(G \cdot x) + \dim(G_x) = \dim(G)$$

for every orbit. We use this to compute the dimension of $X_{n_1+\cdots+n_k}$. Let x be the block diagonal matrix with blocks $J_{n_1}, \ldots, J_{n_k}$. Then k^n together with the endomorphism x can be identified with the $k[t]$-module

$$k[t]/(t^{n_1}) \oplus \cdots \oplus k[t]/(t^{n_k}),$$

on which x acts as multiplication by t. The stabilizer of x for the conjugation action is the centralizer $G_x = \mathrm{Cent}_{\mathrm{GL}_n}(x)$. It is an open subset of the centralizer in $\mathrm{M}_n(k)$, which is the k-linear subspace

$$\begin{aligned} \mathrm{Cent}_{\mathrm{M}_n(k)}(x) &= \mathrm{End}_{k[t]}\big(k[t]/(t^{n_1}) \oplus \cdots \oplus k[t]/(t^{n_k})\big) \\ &= \bigoplus_{i,j=1}^{k} \mathrm{Hom}_{k[t]}\big(k[t]/(t^{n_i}), k[t]/(t^{n_j})\big). \end{aligned}$$

A $k[t]$-linear map $k[t]/(t^a) \to k[t]/(t^b)$ is determined by the image of 1, which can be any element that is killed by t^a. So $\mathrm{Hom}_{k[t]}(k[t]/(t^a), k[t]/(t^b)) = t^{\max\{b-a,0\}}k[t]/(t^b)$ has dimension $\min\{a, b\}$, and we find

$$\dim \mathrm{Cent}_{\mathrm{M}_n(k)}(x) = \sum_{i,j=1}^{k} \min\{n_i, n_j\}.$$

Being a linear subspace, $\mathrm{Cent}_{\mathrm{M}_n(k)}(x)$ is isomorphic to $\mathbb{A}^d$ for d its dimension, and in particular irreducible. The open subset $G_x = \mathrm{Cent}_{\mathrm{M}_n(k)}(x) \cap \mathrm{GL}_n$ is non-empty because

it contains $\mathbb{1}_n$, so it is dense and has the same dimension d. We conclude that

$$\dim X_{n_1+\cdots+n_k} = n^2 - \sum_{i,j=1}^{k} \min\{n_i, n_j\}.$$

Refinement of partitions. By Proposition 4.7 (c), the closure of an orbit is a union of orbits of smaller dimension. Which orbits occur in the closure of $X_{n_1+\cdots+n_k}$? There is a first family of closure relations that comes from refining partitions. The complete answer is more complicated and will be given in the next section.

Let $m = m_1 + m_2$ with $m_1, m_2 \geq 1$ and consider, for $t \in k$, the matrix

$$J_t = \left[\begin{array}{cc|cc} 0 & 1 & & \\ & \ddots & 1 & \\ & & 0 & t \\ \hline & & 0 & 1 \\ & & & \ddots \; 1 \\ & & & 0 \end{array}\right] \begin{matrix} \\ \\ m_1 \\ \\ \\ m_2 \end{matrix} \in \mathrm{M}_m(k),$$

$$\underbrace{\qquad\qquad}_{m}$$

which is the block diagonal matrix with blocks J_{m_1} and J_{m_2}, except that the entry in position (m_1, m_1+1) is t instead of 0. The subset $Z := \{J_t \mid t \in k\} \subseteq \mathrm{M}_m(k)$ is closed and isomorphic to $\mathbb{A}^1$, hence irreducible. For $t \neq 0$, the matrix is conjugate to J_m, but for $t = 0$ its Jordan normal form is $\mathrm{diag}(J_{m_1}, J_{m_2})$. Taking $n = m$ for a moment, we have found an irreducible closed $Z \subseteq X$ such that both $Z \cap X_m$ and $Z \cap X_{m_1+m_2}$ are non-empty, where $Z \cap X_m$ is open. This shows that $X_{m_1+m_2} \subseteq \overline{X_m}$.

The same argument applies to the individual blocks of a larger block diagonal matrix. Given a partition $n = n_1 + \cdots + n_k$ and a *refinement* $n = n'_1 + \ldots + n'_{k'}$, meaning the second partition arises from the first by splitting the $n_1, \ldots, n_k$ into smaller parts, we obtain

$$X_{n'_1+\cdots+n'_{k'}} \subseteq \overline{X_{n_1+\cdots+n_k}}.$$

This gives some closure relations, but not all of them:

Example 5.1. Let $n = 4$. Then $X_{2+2} \subseteq \overline{X_{3+1}}$ as we will see below, but $2 + 2$ does not refine $3 + 1$.

We remark that the partial ordering on partitions by refinement is combinatorially complicated. There is no simple algorithm for determining whether one partition refines another.

The full picture. In order to describe all closure relations, we begin by encoding partitions differently. There is a bijection

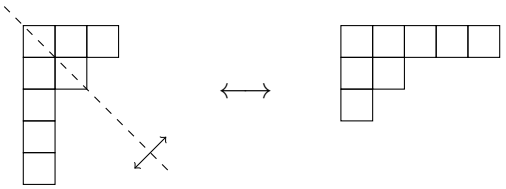
$$\{\text{partitions of } n\} \xrightarrow{\sim} \left\{(\mu_1, \ldots, \mu_n) \in \mathbb{Z}^n \;\middle|\; \mu_1 \geq \cdots \geq \mu_n \geq 0,\ \sum_{i=1}^{n} \mu_i = n\right\} \tag{5.1}$$

given by

$$n = n_1 + \cdots + n_k \longmapsto (\mu_1, \ldots, \mu_n), \qquad \mu_i := \#\{j \mid n_j \geq i\}.$$

Here, # denotes the number of elements of a set. In terms of Young diagrams, μ is the transpose of the partition $n_1 + \cdots + n_k$, with zeros appended to obtain a tuple of length

n:



$$8 = 3 + 2 + 1 + 1 + 1 \qquad \mu = (5, 2, 1, 0, 0, 0, 0, 0)$$

The relevance of this encoding for us is the following. Let $x \in X_{n_1+\cdots+n_k}$ and identify k^n with $k[t]/(t^{n_1}) \oplus \cdots \oplus k[t]/(t^{n_k})$ as before. Then

$$\begin{aligned} \mu_1 &= k = \dim \ker(x), \\ \mu_1 + \mu_2 &= k + \#\{j \mid n_j \geq 2\} = \dim \ker(x^2), \\ &\vdots \\ \mu_1 + \cdots + \mu_n &= n = \dim \ker(x^n). \end{aligned}$$

This leads to the following definitions.

Definition 5.2. (1) For $x \in X$, we define $\mu(x) \in \mathbb{Z}^n$ by

$$\mu(x)_i := \dim \ker(x^i) - \dim \ker(x^{i-1}), \qquad i = 1, \ldots, n.$$

In terms of (5.1), $\mu(x)$ is the tuple encoding the Jordan type of x. Given μ, we write

$$X_\mu := \{x \in X \mid \mu(x) = \mu\}$$

for the corresponding orbit.

(2) The *dominance order* on $\mathbb{Z}^n$ is defined by $\nu \geq \mu$ if

$$\begin{cases} \nu_1 + \cdots + \nu_i \geq \mu_1 + \cdots + \mu_i & \text{for all } i = 1, \ldots, n-1 \\ \nu_1 + \cdots + \nu_n = \mu_1 + \cdots + \mu_n. \end{cases}$$

The second condition will not play a role today because all our tuples sum up to n. A tuple $\mu \in \mathbb{Z}^n$ is called *dominant* if $\mu_1 \geq \mu_2 \geq \ldots \geq \mu_n$.

(3) For μ as in (1), we define

$$X_{\geq \mu} := \{x \in X \mid \mu(x) \geq \mu\}.$$

Example 5.3. For $n = 4$,

$$X_{\geq(2,1,1,0)} = X_{(2,1,1,0)} \cup X_{(2,2,0,0)} \cup X_{(3,1,0,0)} \cup \underbrace{X_{(4,0,0,0)}}_{=\{0\}}.$$

In terms of partitions, $X_{(2,1,1,0)} = X_{3+1}$, $X_{(2,2,0,0)} = X_{2+2}$ and $X_{(3,1,0,0)} = X_{2+1+1}$.

Theorem 5.4. *Let μ and ν be dominant with $\sum_i \mu_i = \sum_i \nu_i = n$ and $\mu_n, \nu_n \geq 0$.*

(a) The subset $X_{\geq\mu} \subseteq X$ is closed.

(b) We have $X_\nu \subseteq \overline{X_\mu}$ if and only if $\nu \geq \mu$. In other words, $\overline{X_\mu} = X_{\geq\mu}$.

(c) In particular, $X_{\geq\mu}$ is an irreducible closed subvariety of $\mathrm{M}_n(k)$ which contains X_μ as a dense open subset.

Proof. (a) By definition of $\mu(x)$ and of the dominance order,

$$\begin{aligned} X_{\geq\mu} &= \{x \in X \mid \dim \ker(x^i) \geq \mu_1 + \cdots + \mu_i \text{ for all } i\} \\ &= \{x \in X \mid \mathrm{rk}(x^i) \leq r_i \text{ for all } i\}, \qquad r_i := n - (\mu_1 + \cdots + \mu_i). \end{aligned}$$

The condition $\mathrm{rk}(x^i) \leq r_i$ means that all $(r_i+1)\times(r_i+1)$-minors of x^i vanish, and these minors are polynomials in the entries of x. So $X_{\geq\mu}$ is the intersection of X with finitely many vanishing sets of polynomials, and hence closed.

(*b*) If $X_\nu \subseteq \overline{X_\mu}$, then $X_\nu \subseteq X_{\geq\mu}$ because $X_{\geq\mu}$ is closed by (a) and contains X_μ. Hence $\nu \geq \mu$. For the converse, we first reduce to an explicit simple situation.

Reduction to $\mu+\alpha \geq \mu$. Given $\nu \geq \mu$, we can find a sequence of elements $\alpha_1, \ldots, \alpha_k$ of the form

$$\alpha_j = (0, \ldots, 0, \overset{a_j}{1}, 0, \ldots, 0, \overset{b_j}{-1}, 0, \ldots, 0), \qquad a_j < b_j$$

such that every intermediate sum $\mu + \alpha_1 + \ldots + \alpha_j$ is dominant and such that

$$\nu = \mu + \alpha_1 + \ldots + \alpha_k.$$

For example, how would one find α_1? Start by comparing μ and ν entry by entry. Take a_1 as the *smallest* index such that $\nu_{a_1} > \mu_{a_1}$, and b_1 as the *largest* such that $\nu_{b_1} < \mu_{b_1}$. Note that in particular

$$\mu_1 = \nu_1, \ldots, \mu_{a_1-1} = \nu_{a_1-1}, \qquad \mu_{b_1+1} = \nu_{b_1+1}, \ldots, \mu_n = \nu_n.$$

So

$$\alpha_1 = (0, \ldots, 0, \overset{a_1}{1}, 0, \ldots, 0, \overset{b_1}{-1}, 0, \ldots, 0)$$

is such that $\mu+\alpha_1$ is still dominant. Iterating this definition starting from $\mu+\alpha_1$ constructs the whole sequence $\alpha_1, \ldots, \alpha_k$. After this reduction, it is left to show the following claim.

Claim. Let μ be dominant and let $\alpha = e_a - e_b$ with $a < b$ be such that $\mu + \alpha$ is again dominant. Then $X_{\mu+\alpha} \subseteq \overline{X_\mu}$.

Proof of the claim. Consider $x \in X_\mu$. The dominance of $\mu+\alpha$ means that $\mu_{a-1} > \mu_a$ (if $a > 1$) and $\mu_b > \mu_{b+1}$. In terms of the partition, this says that x has a Jordan block of size exactly $a-1$ (if $a > 1$) and a Jordan block of size exactly b. Replacing these two blocks by blocks of sizes a and $b-1$ changes μ into $\mu+\alpha$. (If $a = 1$, this means that a block of size b is replaced by two blocks of sizes 1 and $b-1$.) So it suffices to consider the $\mathbb{A}^1$-family of $(a+b-1)\times(a+b-1)$-matrices

$$x_t = \left(\begin{array}{ccc|ccc} 0 & 1 & & & & \\ & \ddots & \ddots & & & \\ & & 1 & & & \\ & & 0 & & & t \\ \hline & & & 0 & 1 & \\ & & & & \ddots & 1 \\ & & & & & 0 \end{array}\right) \begin{array}{l} \left.\vphantom{\begin{array}{c}a\\a\\a\end{array}}\right\} b-1 \\ \left.\vphantom{\begin{array}{c}a\\a\\a\end{array}}\right\} a \end{array} \qquad t \in k,$$

which are block diagonal with blocks J_{b-1} and J_a, except for the entry t in position $(b-1, a+b-1)$. For $t = 0$, the matrix x_0 has Jordan blocks of sizes $b-1$ and a. For $t \neq 0$, one computes

$$\dim\ker(x_t^i) = \begin{cases} 2i, & \text{if } 1 \leq i \leq a-1, \\ i+a-1, & \text{if } a \leq i \leq b, \end{cases}$$

which is the kernel dimension pattern of a matrix with Jordan blocks of sizes $a-1$ and b. So after adding the remaining Jordan blocks of x, we have $x_t \in X_\mu$ for $t \neq 0$ and $x_0 \in X_{\mu+\alpha}$. As in the previous section, $\{x_t \mid t \in k\} \cong \mathbb{A}^1$ is irreducible, so $x_0 \in \overline{\{x_t \mid t \neq 0\}} \subseteq \overline{X_\mu}$, and hence $X_{\mu+\alpha} \subseteq \overline{X_\mu}$ by GL_n-stability of the closure.

(c) The orbit X_μ is the image of the irreducible variety GL_n and hence irreducible, so its closure $X_{\geq\mu}$ is irreducible as well. Moreover,

$$X_\mu = X_{\geq\mu} \setminus \bigcup_{\nu>\mu} X_{\geq\nu}$$

is the complement of a finite union of closed subsets and hence open in $X_{\geq\mu}$. □

5.2. **Case study 2: $\mathrm{Hom}(\mathrm{GL}_n, \mathbb{G}_a)$.** The following is closely related to conjugation in GL_n and the construction of Δ in (4.5).

Example 5.5. We claim that there are no non-zero homomorphisms $\mathrm{GL}_n \to \mathbb{G}_a$. That is, $\mathrm{Hom}(\mathrm{GL}_n, \mathbb{G}_a) = 0$.

Proof of claim. First note that $\mathrm{Hom}(\mathbb{G}_m, \mathbb{G}_a) = 0$. Namely, by looking at coordinate rings, we get

$$\mathrm{Hom}(\mathbb{G}_m, \mathbb{G}_a) = \{a(t) \in k[t, t^{-1}] : a(xy) = a(x) + a(y)\}$$

and this is just $\{0\}$. Second, we consider the diagonal torus $T = \mathbb{G}_m^n \subset \mathrm{GL}_n$. The above shows that every $\varphi : \mathrm{GL}_n \to \mathbb{G}_a$ satisfies $\varphi|_T = 0$. Moreover, $\mathbb{G}_a$ abelian, so $\varphi(xyx^{-1}) = \varphi(y)$ for all $x, y \in \mathrm{GL}_n$. Thus if y is diagonalizable, then $\varphi(y) = 0$.

Now we observe that the diagonalizable elements are dense in GL_n:

$$\{\text{diagonalizable } x \text{ with different eigenvalues}\} \subset \{\text{diagonalizable } x\} \subset \mathrm{GL}_n$$

and the left-most set is the principal open $D(\Delta)$ with Δ the discriminant. We conclude that $\varphi = 0$ on all of GL_n.

6. Projective varieties and quotients (10/14/2026)

Recall our current goal from last lecture: Given a LAG G and a closed subgroup $H \subseteq G$, we want to endow the set of cosets G/H with a natural variety structure. Our plan was to construct a variety X with G-action and a point $x \in X$ whose stabilizer is H, and to then identify G/H with the orbit $G \cdot x$. The varieties X we will use for this are *projective*, so the first half of this lecture is an introduction to projective varieties. In the second half, we carry out the construction of G/H as variety.

6.1. **Projective varieties.** Reference: [1, §1.6]

Definition 6.1. The *projective n-space* $\mathbb{P}^n$ over k is the set of lines in k^{n+1}, i.e.

$$\mathbb{P}^n := k^\times \backslash \big(k^{n+1} \setminus \{0\}\big).$$

We write $[x_0 : \ldots : x_n]$ for the point of $\mathbb{P}^n$ defined by a vector $(x_0, \ldots, x_n) \neq 0$. More generally, for a finite-dimensional k-vector space V, we denote by $\mathbb{P}(V)$ the set of lines in V.

Topology. Let $f \in k[X_0, \ldots, X_n]$ be a homogeneous polynomial. Whether or not $f(x_0, \ldots, x_n)$ vanishes only depends on the line $[x_0 : \ldots : x_n]$, because $f(\lambda x) = \lambda^{\deg(f)} f(x)$ for $\lambda \in k^\times$. So we may define the vanishing locus of f and its complement,

$$V(f) := \{[x_0 : \ldots : x_n] \in \mathbb{P}^n \mid f(x_0, \ldots, x_n) = 0\}, \qquad D(f) := \mathbb{P}^n \setminus V(f).$$

The *Zariski topology* on $\mathbb{P}^n$ is the topology generated by the sets $D(f)$, for f ranging over all homogeneous polynomials. The $D(f)$ form a basis for this topology. The closed subsets are precisely the sets of the form $V(I) = \bigcap_f V(f)$, where $I \subseteq k[X_0, \ldots, X_n]$ is a homogeneous ideal and f runs through its homogeneous elements.

Theorem 6.2 (Hilbert)**.** *There is a bijection*

$$\left\{ \begin{array}{c} \text{radical homogeneous ideals} \\ I \subseteq k[X_0,\dots,X_n] \text{ with } I \neq (X_0,\dots,X_n) \end{array} \right\} \overset{1:1}{\longleftrightarrow} \{\text{closed } Z \subseteq \mathbb{P}^n\}$$
$$I \longmapsto V(I)$$
$$I(Z) \longmapsfrom Z.$$

Here, $I(Z)$ denotes the ideal generated by all homogeneous polynomials that vanish on Z.

The ideal $(X_0,\dots,X_n)$ has to be excluded because $V((X_0,\dots,X_n)) = \varnothing = V((1))$.

Variety structure. Let f be a non-zero homogeneous polynomial. We define the ring of regular functions on $D(f)$ as

$$k[D(f)] := k[X_0,\dots,X_n,f^{-1}]_{\deg 0}.$$

That is, we take the degree 0 elements of the graded ring $k[X_0,\dots,X_n,f^{-1}]$ where the grading is defined by $\deg(p/q) = \deg(p) - \deg(q)$. In other words, a regular function on $D(f)$ is a quotient g/f^r with g homogeneous of degree $r\deg(f)$. Such a quotient indeed defines a function $D(f) \to k$, because numerator and denominator are homogeneous of the same degree. Then $k[D(f)]$ is a reduced finite type k-algebra with $\mathrm{Sp}(k[D(f)]) = D(f)$, and this defines a variety structure on $\mathbb{P}^n$ in the sense of §3.2.

Example 6.3. Consider $\mathbb{P}^2$ with coordinates $[x:y:z]$. The quotient $(y^2+z^2)/x^2$ defines a regular function on $D(x) \subseteq \mathbb{P}^2$. At the point $[2:1:3]$, it takes the value $(1+9)/4 = 10/4$, and this value does not change if we replace $(2,1,3)$ by any other non-zero vector on the line $\langle(2,1,3)\rangle$.

Question. Let V be a vector space of dimension $n+1$ and let $\ell \in \mathbb{P}(V)$ be a line. How to describe an affine neighborhood of ℓ?

Answer. Choose a complement W of ℓ, meaning a subspace $W \subseteq V$ with $V = \ell \oplus W$. Choose a basis vector e_0 of ℓ and a basis $e_1,\dots,e_n$ of W. The lines $\ell' \subseteq V$ that are not contained in W are precisely the lines of the form

$$\ell' = k \cdot \Big(e_0 + \sum_{i=1}^{n} t_i e_i\Big)$$

for unique $t_1,\dots,t_n \in k$. This defines a bijection

$$k^n = \mathrm{Sp}(k[T_1,\dots,T_n]) \xrightarrow{\sim} \{\ell' \in \mathbb{P}(V) \mid \ell' \not\subseteq W\} = \mathbb{P}(V) \setminus \mathbb{P}(W).$$

In the coordinates $[x_0 : \dots : x_n]$ defined by the basis $e_0,\dots,e_n$, the right hand side is the principal open subset $D(X_0)$, and the above bijection is the isomorphism

$$\mathrm{Sp}\big(k[X_1/X_0,\dots,X_n/X_0]\big) \xrightarrow{\sim} D(X_0).$$

In particular, $\mathbb{P}(V) \setminus \mathbb{P}(W)$ is an affine open neighborhood of ℓ.

Example 6.4. Sometimes a morphism is directly given in projective coordinates. For example, consider

$$\mathbb{P}^1 \longrightarrow V(XY - Z^2) \subseteq \mathbb{P}^2, \qquad [a:b] \longmapsto [a^2 : b^2 : ab].$$

This is well-defined because a^2, b^2 and ab are homogeneous of the same degree and have no common zero on $k^2 \setminus \{0\}$, and the image lies in $V(XY - Z^2)$ because $a^2b^2 = (ab)^2$. The morphism property can be checked in charts. The preimage of $D(X)$ is $D(a)$, and on this chart the map is given by

$$\mathbb{A}^1 \cong D(a) \longrightarrow D(X) \cong \mathbb{A}^2$$
$$[1:b] \longmapsto [1 : b^2 : b],$$

i.e. by the polynomial map $b \mapsto (b^2, b)$ in the affine coordinate b on $D(a)$ and the affine coordinates $(Y/X, Z/X)$ on $D(X)$. The chart $D(b)$ is treated in the same way.

Here is an observation about this example. In affine coordinates, the map on the chart $D(a)$ is the isomorphism

$$k \xrightarrow{\sim} V(y - z^2) \subseteq k^2, \qquad b \longmapsto (b^2, b),$$

onto the closed subvariety $V(y - z^2) = V(XY - Z^2) \cap D(X)$. The same happens on the chart $D(b)$. Since $V(XY - Z^2) \subseteq D(X) \cup D(Y)$, it follows that

$$\mathbb{P}^1 \xrightarrow{\sim} V(XY - Z^2)$$

is an isomorphism.

Example 6.5 (Veronese embedding)**.** The previous example generalizes to the *Veronese embedding*

$$\mathbb{P}^n \hookrightarrow \mathbb{P}^{\binom{n+d}{d}-1}$$

$$[x_0 : \ldots : x_n] \longmapsto [x^I]_{|I|=d},$$

where $I = (a_0, \ldots, a_n)$ runs over all exponent vectors with $|I| := a_0 + \ldots + a_n = d$, and where $x^I := x_0^{a_0} \cdots x_n^{a_n}$. The example above is the case $n = 1$ and $d = 2$. The Veronese embedding is a closed immersion.

Definition 6.6. A *projective variety* is a variety that is isomorphic to a closed subvariety of some $\mathbb{P}^n$. A *quasi-projective variety* is a variety that is isomorphic to a locally closed subvariety of some $\mathbb{P}^n$.

Example 6.7. Every affine variety is isomorphic to a closed subvariety of some $\mathbb{A}^n = D(X_0) \subset \mathbb{P}^n$. Since $D(X_0)$ is open in $\mathbb{P}^n$, affine varieties are hence quasi-projective.

6.2. **Products.** Reference: [1, §1.7]

Recall that the product of two affine varieties $X = \mathrm{Sp}(A)$ and $Y = \mathrm{Sp}(B)$ is $X \times Y = \mathrm{Sp}(A \otimes_k B)$, and that the fibre product of $Y = \mathrm{Sp}(B)$ and $Z = \mathrm{Sp}(C)$ over $X = \mathrm{Sp}(A)$ is $Y \times_X Z = \mathrm{Sp}((B \otimes_A C)_{\mathrm{red}})$. We now extend these constructions to general varieties.

Let X and Y be varieties with finite affine open coverings $X = \bigcup_i U_i$ and $Y = \bigcup_j V_j$. The product $X \times Y$ is the product of the underlying sets, endowed with the variety structure for which the subsets $U_i \times V_j$ are affine open subvarieties. One checks that this is well-defined, i.e. independent of the choice of the coverings, and that $X \times Y$ has the universal property of a product in the category of varieties over k.

Similarly, let S be a variety and let $f : X \to S$ and $g : Y \to S$ be morphisms,

$$\begin{array}{ccc} X & & Y. \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle g} & \\ & S & \end{array}$$

The fibre product $X \times_S Y$ is then constructed as follows.

(1) Cover S by affine open subsets, $S = \bigcup_k S_k$.

(2) Cover their preimages by affine open subsets,

$$f^{-1}(S_k) = \bigcup_i U_{ki}, \qquad g^{-1}(S_k) = \bigcup_j V_{kj}.$$

(3) Then $X \times_S Y = \{(x, y) \in X \times Y \mid f(x) = g(y)\}$ is endowed with the variety structure for which

$$X \times_S Y = \bigcup_k \bigcup_{i,j} U_{ki} \times_{S_k} V_{kj}$$

is an affine open covering.

Definition 6.8. The *Segre embedding* is the morphism

$$\mathbb{P}^n \times \mathbb{P}^m \hookrightarrow \mathbb{P}^{nm+n+m}$$
$$([x_0 : \ldots : x_n], [y_0 : \ldots : y_m]) \longmapsto [x_i y_j]_{i,j}.$$

One checks that it is a closed immersion.

Example 6.9. The product of two projective varieties is projective. Namely, if $X \subseteq \mathbb{P}^m$ and $Y \subseteq \mathbb{P}^n$ are closed subvarieties, then

$$X \times Y \underset{\text{closed}}{\subseteq} \mathbb{P}^m \times \mathbb{P}^n \hookrightarrow \mathbb{P}^{mn+m+n}$$

is closed, where the second arrow is the Segre embedding.

6.3. **Grassmannians.** Reference: [1, §1.8].

Let V be an n-dimensional k-vector space and let $0 \leq d \leq n$. The *Grassmannian* of d-dimensional subspaces of V is the set

$$\mathrm{Gr}_d(V) := \{W \subseteq V \mid W \text{ is a } d\text{-dimensional subspace}\}.$$

For example, $\mathrm{Gr}_1(V) = \mathbb{P}(V)$. We make $\mathrm{Gr}_d(V)$ into a variety by specifying an affine open covering, similar to as we did for $\mathbb{P}(V)$ before. Given $W \in \mathrm{Gr}_d(V)$, choose a complement U of W, so $V = W \oplus U$. We declare the subset

$$\{W' \in \mathrm{Gr}_d(V) \mid W' \cap U = \{0\}\}$$

to be open. Choose a basis $e_1, \ldots, e_d$ of W and a basis $e_{d+1}, \ldots, e_n$ of U. Every W' with $W' \cap U = \{0\}$ has a unique basis of the form

$$e_i + \sum_{j=d+1}^{n} t_{ij} e_j, \qquad i = 1, \ldots, d,$$

with coefficients $t_{ij} \in k$. Namely, $W' \cap U = \{0\}$ means that the projection $W' \to V/U \cong W$ is an isomorphism, and the above basis is the preimage of $e_1, \ldots, e_d$. We use the coefficients t_{ij} to identify

$$k^{d(n-d)} = \mathrm{Sp}\big(k[T_{ij} \mid 1 \leq i \leq d,\ d+1 \leq j \leq n]\big) \xrightarrow{\sim} \{W' \in \mathrm{Gr}_d(V) \mid W' \cap U = \{0\}\}.$$

One checks that this provides a variety structure on $\mathrm{Gr}_d(V)$.

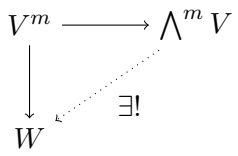
Proposition 6.10. *The Grassmannian* $\mathrm{Gr}_d(V)$ *is projective. That is, there exists a closed immersion* $\mathrm{Gr}_d(V) \hookrightarrow \mathbb{P}^N$ *for some* N.

The embedding will be constructed using exterior powers, so we first recall their definition.

Definition 6.11. Let V be a k-vector space and let $m \geq 0$. The *m-th exterior power* of V is a k-vector space $\bigwedge^m V$ together with an alternating map

$$V^m \longrightarrow \bigwedge^m V, \qquad (v_1, \ldots, v_m) \longmapsto v_1 \wedge \cdots \wedge v_m,$$

that is universal in the following sense: For every alternating map $V^m \to W$, there exists a unique k-linear map $\bigwedge^m V \to W$ such that the diagram



commutes. Here, a map $V^m \to W$ is called *alternating* if it is multilinear and vanishes whenever two of its arguments coincide, i.e. $(\dots, v, \dots, v, \dots) \mapsto 0$ for all $v \in V$. Since the characteristic of k is not 2, this is equivalent to being multilinear and antisymmetric, meaning that $(\dots, v, \dots, w, \dots) = -(\dots, w, \dots, v, \dots)$.

The exterior power exists and is unique up to unique isomorphism by its universal property. We recall some of its properties.

• If $e_1, \dots, e_n$ is a basis of V, then the vectors $e_{i_1} \wedge \cdots \wedge e_{i_m}$ with $i_1 < \cdots < i_m$ form a basis of $\bigwedge^m V$. In particular,

$$\dim \bigwedge^m V = \tbinom{\dim V}{m}.$$

• Functoriality: Every linear map $\varphi : V \to W$ induces a linear map $\bigwedge^m \varphi : \bigwedge^m V \to \bigwedge^m W$ by

$$(\textstyle\bigwedge^m \varphi)(v_1 \wedge \cdots \wedge v_m) = \varphi(v_1) \wedge \cdots \wedge \varphi(v_m).$$

It is obtained by applying the universal property to the alternating map obtained as the composition $V^m \to W^m \to \bigwedge^m W$:

$$\begin{array}{ccc} V^m & \xrightarrow{\varphi^m} & W^m \\ \downarrow & & \downarrow \\ \bigwedge^m V & \xrightarrow{\exists!\ \bigwedge^m \varphi} & \bigwedge^m W. \end{array}$$

• If $W \subseteq V$ is a d-dimensional subspace and $m \le d$, then $\bigwedge^m W \subseteq \bigwedge^m V$ is a subspace of dimension $\binom{d}{m}$. In this way, we obtain a map

$$\begin{aligned} \mathrm{Gr}_d(V) &\longrightarrow \mathrm{Gr}_{\binom{d}{m}}\Big(\textstyle\bigwedge^m V\Big) \\ W &\longmapsto \textstyle\bigwedge^m W. \end{aligned}$$

This map is a closed immersion, which we now verify in the case $m = d$. In this case, $\bigwedge^d W$ is a line, so the target is the projective space $\mathbb{P}(\bigwedge^d V) \cong \mathbb{P}^{\binom{n}{d}-1}$, and the map is called the *Plücker embedding*.

Proof of Proposition 6.10. We show that the Plücker embedding

$$\begin{aligned} \varphi : \mathrm{Gr}_d(V) &\longrightarrow \mathbb{P}\Big(\textstyle\bigwedge^d V\Big) \\ W &\longmapsto \textstyle\bigwedge^d W \end{aligned}$$

is a closed immersion.

φ is injective. Let $W \neq W'$ be two d-dimensional subspaces and put $m := \dim(W \cap W') < d$. Choose a basis $e_1, \dots, e_m$ of $W \cap W'$ and extend it to a basis $e_1, \dots, e_m, e_{m+1}, \dots, e_d$ of W and to a basis $e_1, \dots, e_m, e'_{m+1}, \dots, e'_d$ of W'. Then the vectors $e_1, \dots, e_d, e'_{m+1}, \dots, e'_d$ are linearly independent and can be extended to a basis of V. Hence

$$e_1 \wedge \cdots \wedge e_m \wedge e_{m+1} \wedge \cdots \wedge e_d \qquad \text{and} \qquad e_1 \wedge \cdots \wedge e_m \wedge e'_{m+1} \wedge \cdots \wedge e'_d$$

are two different elements of a basis of $\bigwedge^d V$, and in particular they span different lines.

φ is locally on $\mathrm{Gr}_d(V)$ a closed immersion. Fix a basis $e_1, \dots, e_n$ of V and consider the chart of $\mathrm{Gr}_d(V)$ defined by $W = \langle e_1, \dots, e_d \rangle$ and $U = \langle e_{d+1}, \dots, e_n \rangle$. For W' in this chart

with basis $e_i + \sum_{j>d} t_{ij} e_j$ as before, we compute

$$\Big(e_1 + \sum_{j=d+1}^{n} t_{1j} e_j\Big) \wedge \cdots \wedge \Big(e_d + \sum_{j=d+1}^{n} t_{dj} e_j\Big)$$
$$= e_1 \wedge \cdots \wedge e_d + \sum_{i=1}^{d} \sum_{j=d+1}^{n} t_{ij}\, e_1 \wedge \cdots \wedge e_{i-1} \wedge e_j \wedge e_{i+1} \wedge \cdots \wedge e_d$$
$$+ \big(\text{terms of degree} \geq 2 \text{ in the } t_{ij}\big).$$

The terms of degree ≥ 2 in the t_{ij} are linear combinations of basis vectors $e_{i_1} \wedge \cdots \wedge e_{i_d}$ with at least two indices $> d$. For $I = \{i_1 < \cdots < i_d\}$, let X_I denote the coordinate on $\mathbb{P}(\bigwedge^d V)$ corresponding to the basis vector $e_{i_1} \wedge \cdots \wedge e_{i_d}$. The computation shows that the chart maps into the principal open subset $D(X_{\{1,\ldots,d\}})$ and we show that

$$k^{d(n-d)} \longrightarrow D(X_{\{1,\ldots,d\}}) \cong k^{\binom{n}{d}-1}$$

is a closed immersion. On coordinate rings, this map is given by a ring map

$$k[T_{ij}] \xleftarrow{\varphi^*} k\big[X_I/X_{\{1,\ldots,d\}} \;\big|\; I \neq \{1,\ldots,d\}\big].$$

For $I = (\{1,\ldots,d\} \setminus \{i\}) \cup \{j\}$ with $j > d$, we have

$$e_1 \wedge \cdots \wedge e_{i-1} \wedge e_j \wedge e_{i+1} \wedge \cdots \wedge e_d = (-1)^{d-i}\, e_1 \wedge \cdots \wedge \widehat{e_i} \wedge \cdots \wedge e_d \wedge e_j,$$

because e_j has to be swapped past the $d-i$ vectors $e_{i+1}, \ldots, e_d$. By the above expansion, this means $\varphi^*(X_I/X_{\{1,\ldots,d\}}) = (-1)^{d-i} T_{ij}$. So every T_{ij} lies in the image of φ^*, meaning that φ^* is surjective. This precisely says that φ restricted to the chart $\{W' \cap U = \{0\}\}$ is a closed immersion into $D(X_{\{1,\ldots,d\}})$.

End of proof. A small argument (exercise) shows that in fact $\varphi^{-1}(D(X_{1,\ldots,d})) = \{W' \cap U = \{0\}\}$, so we have shown that φ is an isomorphism onto a locally closed subset of $\mathbb{P}(\wedge^d V)$. After proving that $\mathrm{Im}(\varphi)$ is closed (exercise), we get that φ is a closed immersion. □

6.4. **Constructing quotients as varieties.** Reference: [1, §11].

The group $\mathrm{GL}(V)$ acts on $\mathrm{Gr}_d(V)$ by $g \cdot W := g(W)$. We admit that this action comes from a morphism $\mathrm{GL}(V) \times \mathrm{Gr}_d(V) \to \mathrm{Gr}_d(V)$. It is clearly transitive. The stabilizer of $\langle e_1, \ldots, e_d \rangle \subseteq k^n$ in GL_n is the group of block upper triangular matrices, so we have constructed the new quotients

$$\mathrm{GL}_n \Big\backslash \left\{ \begin{pmatrix} \star_d & \star \\ 0 & \star_{n-d} \end{pmatrix} \right\} \xrightarrow{\sim} \mathrm{Gr}_d(k^n)$$
$$g \longmapsto g \cdot \langle e_1, \ldots, e_d \rangle.$$

The projective setting also contains our earlier affine examples. For instance, the subgroup

$$\mathrm{GL}_n \cong \left\{ \begin{pmatrix} 1 & \\ & \star_n \end{pmatrix} \right\} \subseteq \mathrm{GL}_{n+1}$$

preserves the open subset $D(X_0) \cong k^n$ of $\mathbb{P}^n$ and acts on it in the standard way.

We can now carry out the first step of our plan.

Proposition 6.12. *Let G be a LAG and let $H \subseteq G$ be a closed subgroup. Then there exist a projective variety X with G-action and a point $x \in X$ such that $H = G_x$ is the stabilizer of x.*

Throughout the proof, we use the right translation action of G on $k[G]$,

$$(\rho_x f)(y) := f(yx), \qquad x, y \in G,\ f \in k[G].$$

Since $H \subseteq G$ is a closed subvariety, we have $k[H] = k[G]/I$ for the radical ideal $I = I(H)$ of functions vanishing on H.

Lemma 6.13. *We have* $H = \{x \in G \mid \rho_x(I) = I\}$.

Proof. Let $x \in H$ and $f \in I$. For every $y \in H$, we have $yx \in H$ and hence $(\rho_x f)(y) = f(yx) = 0$. This shows $\rho_x(I) \subseteq I$. Applying the same argument to x^{-1} gives $\rho_{x^{-1}}(I) \subseteq I$, and hence $\rho_x(I) = I$.

Conversely, let $x \in G$ with $\rho_x(I) = I$. Then for every $f \in I$, the function $\rho_x f$ again lies in I and hence vanishes at $e \in H$. So $f(x) = f(ex) = (\rho_x f)(e) = 0$ for all $f \in I$, which means $x \in V(I) = H$. □

Proof of Proposition 6.12. Let $f_1, \dots, f_r$ be ideal generators of I. Choose a finite-dimensional subspace $W \subseteq k[G]$ that contains $f_1, \dots, f_r$ and that is stable under all ρ_x, $x \in G$. Such W exists by our proof of Theorem 3.2. Put $M := W \cap I$ and $d := \dim(M)$. Then M is a d-dimensional subspace of W which generates I as an ideal. We claim that

$$H = \{x \in G \mid \rho_x(M) = M\}. \tag{6.1}$$

Indeed, if $x \in H$, then ρ_x preserves both W and I by Lemma 6.13, hence also M. Conversely, if $\rho_x(M) = M$, then

$$\rho_x(I) = \rho_x(k[G] \cdot M) = \rho_x(k[G]) \cdot \rho_x(M) = k[G] \cdot M = I$$

because ρ_x is a ring automorphism, so $x \in H$ by Lemma 6.13.

Now let G act on the projective variety $X := \mathrm{Gr}_d(W)$ through $\rho : G \to \mathrm{GL}(W)$ and the standard action of $\mathrm{GL}(W)$ on $\mathrm{Gr}_d(W)$. Then (6.1) precisely says that H is the stabilizer of the point $x := M \in \mathrm{Gr}_d(W)$, finishing the proof of Proposition 6.12. □

Remark 6.14. If desired, one can even achieve $X = \mathbb{P}^N$ for some N. Namely, composing with the Plücker embedding, G acts on $\mathbb{P}(\bigwedge^d W)$ through $\mathrm{GL}(W)$, and the stabilizer of the point $\bigwedge^d M \in \mathbb{P}(\bigwedge^d W)$ is still H, because the Plücker embedding is injective and $\mathrm{GL}(W)$-equivariant.

6.5. **The quotient property.** Reference: [1, §12].

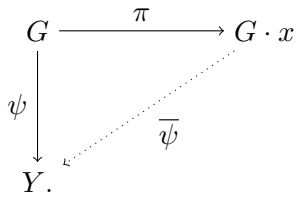
Given a closed subgroup $H \subseteq G$, Proposition 6.12 provides a projective variety X with G-action and a point $x \in X$ with $H = G_x$. The orbit $G \cdot x \subseteq X$ is locally closed by Proposition 4.7, hence a quasi-projective variety, and the orbit map induces a bijection

$$G/H \xrightarrow{\sim} G \cdot x, \qquad gH \longmapsto gx.$$

It remains to show that this bijection is a geometric quotient for the right action of H on G:

Theorem 6.15. *The orbit map $\pi : G \to G \cdot x$, $g \mapsto gx$, is a categorical quotient for the action of $H = G_x$ on G. Since π induces a bijection $G/H \xrightarrow{\sim} G \cdot x$, it is even a geometric quotient.*

Proof. Clearly, π is H-invariant. Let $\psi : G \to Y$ be an H-invariant morphism. We have to show that there exists a unique morphism $\overline{\psi} : G \cdot x \to Y$ with $\psi = \overline{\psi} \circ \pi$:



Uniqueness is clear, since the only possibility is $\overline{\psi}(gx) = \psi(g)$. We want to show that such defined $\overline{\psi}$ is, in fact, a morphism.

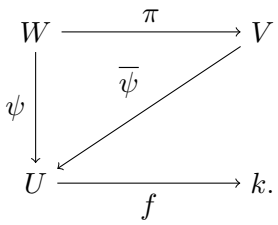
$\overline{\psi}$ is continuous. We use a non-trivial result from algebraic geometry: Flat morphisms of varieties are open [2, Tags 01UA and 00I1]. That is, if $\varphi : A \to B$ is a flat map of reduced finite type k-algebras, then the map $\mathrm{Sp}(\varphi) : \mathrm{Sp}(B) \to \mathrm{Sp}(A)$ is open with respect to the Zariski topology. In our setting, we know that $\pi : G \to G \cdot x$ is smooth, see the argument around (4.3). Smooth morphisms are also flat, so the cited result applies, and we get that π is open.

Now let $U \subseteq Y$ be open. Since π is surjective and $\psi = \overline{\psi} \circ \pi$, we have

$$\overline{\psi}^{-1}(U) = \pi\big(\psi^{-1}(U)\big).$$

The set $\psi^{-1}(U)$ is open by continuity of ψ, so its image $\pi(\psi^{-1}(U))$ is open by openness of π. We conclude that $\overline{\psi}$ is continuous.

$\overline{\psi}$ is a morphism. Let $U \subseteq Y$ be an open subset and let $f \in \mathscr{O}_Y(U)$ be a regular function. We need to show that $f \circ \overline{\psi}$ is regular on $V := \overline{\psi}^{-1}(U)$. Put $W := \psi^{-1}(U) = \pi^{-1}(V)$. The situation is as follows:



We know that $f \circ \psi = f \circ \overline{\psi} \circ \pi$ is regular on W, and we want to deduce that $f \circ \overline{\psi}$ is regular on V. This is precisely the content of the next lemma, which completes the proof of the theorem. □

Lemma 6.16. *Let $V \subseteq G \cdot x$ be open and let $h : V \to k$ be a function such that $h \circ \pi : \pi^{-1}(V) \to k$ is regular. Then h is regular.*

Proof. Regularity is a local property. So by covering V by affine open subsets, we may assume that V itself is affine. Since $G \cdot x$ is locally closed in the projective variety X it is separated. G is affine, so it is automatically true that $W := \pi^{-1}(V) \subset G$ is again affine. This is a fact we will simply admit.

Write $A := k[V]$ and $B := k[W]$, so that $\pi : W \to V$ induces an injective pullback map $\pi^* : A \to B$ because π is surjective.

By assumption, $b := h \circ \pi$ is an element of B. It is constant on the fibres of π, which are cosets gH. We have to show that b lies in $\pi^*(A)$. Namely, if $b = \pi^*(a) = a \circ \pi$ for some $a \in A$, then $h = a$ by surjectivity of π, so h is regular.

The main tool for this is *faithfully flat descent*.

Theorem 6.17 ([2, Tag 023M]). *Let $A \to B$ be a faithfully flat ring homomorphism. Then*

$$A = \{b \in B \mid b \otimes 1 = 1 \otimes b \text{ as elements of } B \otimes_A B\}.$$

Recall that faithfully flat means flat and surjective on spectra. We have seen above that π is flat, and it is surjective, so $A \to B$ is faithfully flat. We even know that $A \to B$ is smooth. Being smooth is preserved under tensor products and being smooth over a reduced ring implies being reduced, so we get that $B \otimes_A B$ is reduced. It follows that the fiber product $W \times_V W$ equals $\mathrm{Sp}(B \otimes_A B)$ on the nose. (No need to pass to $(B \otimes_A B)_{\mathrm{red}}$.)

$$\mathrm{Sp}(B \otimes_A B) = W \times_V W = \{(w, w') \in W \times W \mid \pi(w) = \pi(w')\}.$$

The condition $b \otimes 1 = 1 \otimes b$ means that $b \circ p_1 = b \circ p_2$ for the two projections $p_1, p_2 : W \times_V W \to W$. Theorem 6.17 gives that b already lies in A, which is precisely the claim of the lemma. □

Remark 6.18. We want to stay focused on linear algebraic groups in this course which is why we have black-boxed the openness of π and Theorem 6.17 in the above proof. Both are important results in algebraic geometry though, and it is worth appreciating how they allowed us to understand properties of $G \to G \cdot x$.

Summary. Let G act on a variety X and let $x \in X$. Then the orbit $G \cdot x \subseteq X$ is locally closed, hence a variety, and the orbit map $\pi : G \to G \cdot x$ is a geometric quotient $G/G_x \xrightarrow{\sim} G \cdot x$. Moreover, our proof has shown the following two further interesting properties.

(1) The topology of $G \cdot x$ is the quotient topology, i.e. $V \subseteq G \cdot x$ is open if and only if $\pi^{-1}(V)$ is open. This follows from π being surjective and open.

(2) For every affine open subset $V \subseteq G \cdot x$, the preimage $\pi^{-1}(V)$ is affine and

$$k[V] = k[\pi^{-1}(V)]^{G_x}$$

is the ring of regular functions on $\pi^{-1}(V)$ that are invariant under right translation by G_x. In particular, if $G \cdot x$ is affine, then $G \cdot x = \mathrm{Sp}(k[G]^{G_x})$.

7. The Lie algebra (10/21/2026)

Reference: [1, §9].

7.1. **Motivation.** In the last lecture, we showed that for every closed subgroup $H \subseteq G$, the geometric quotient G/H exists and is a quasi-projective variety.

New question. How to describe the closed subgroups $H \subseteq G$?

The idea is to linearize the problem by passing to the Lie algebra. The following theorem will be next big result.

Theorem 7.1. *There is a faithful functor*

$$\begin{array}{ccc} \{\text{connected LAGs}/k\} & \longrightarrow & \{\text{Lie algebras}/k\} \\ G & \longmapsto & \mathrm{Lie}(G). \end{array}$$

Here, faithful means that for all connected LAGs G_1 and G_2, the map

$$\mathrm{Hom}(G_1, G_2) \hookrightarrow \mathrm{Hom}(\mathrm{Lie}(G_1), \mathrm{Lie}(G_2))$$

is injective. The functor itself is defined for all LAGs, connectedness is only needed for its faithfulness. Today, we construct $\mathrm{Lie}(G)$ as a k-vector space with Lie bracket.

7.2. **Vector fields and tangent spaces.** Recall the following definition from differential geometry. Let M be a smooth manifold. A *smooth vector field on* M is a derivation

$$X \in \mathrm{Der}_{\mathbb{R}}\big(C^\infty(M), C^\infty(M)\big).$$

That is, X is an $\mathbb{R}$-linear map $X : C^\infty(M) \to C^\infty(M)$ such that

(a) $X(c) = 0$ for every constant function c, and
(b) the Leibniz rule $X(fg) = gX(f) + fX(g)$ holds.

A *tangent vector* at a point $x \in M$ is a derivation

$$X_x : C^\infty(M) \longrightarrow \mathbb{R},$$

where the Leibniz rule now reads $X_x(fg) = g(x)X_x(f) + f(x)X_x(g)$. For example, the evaluation $f \mapsto X(f)(x)$ of a vector field X at x is a tangent vector at x.

All these notions are entirely algebraic, so we can directly transfer them to algebraic geometry!

Definition 7.2. Let R be a ring, let A be an R-algebra and let M be an A-module. An *R-derivation* from A to M is an R-linear map $X : A \to M$ that satisfies the Leibniz rule

$$X(fg) = fX(g) + gX(f), \qquad f, g \in A.$$

The set of all R-derivations of A into M is denoted by $\mathrm{Der}_R(A, M)$. It is an A-module.

The Leibniz rule gives $X(1) = X(1 \cdot 1) = 2X(1)$, so $X(1) = 0$ and hence $X(c) = 0$ for all c in the image of R. This is why condition (a) above does not have to be imposed separately.

Definition 7.3. Let $G = \mathrm{Sp}(A)$ be an affine variety, so $A = k[G]$.

(1) A *vector field* on G is a derivation $X \in \mathrm{Der}_k(k[G], k[G])$. We write

$$T(G) := \mathrm{Der}_k(k[G], k[G])$$

for the space of all vector fields on G.

(2) The *tangent space* of G at a point $x \in G$ is

$$T_x(G) := \mathrm{Der}_k\big(k[G], k[G]/\mathfrak{m}_x\big),$$

where $k[G]/\mathfrak{m}_x$ is just k viewed as a $k[G]$-module via $f \mapsto f(x)$.

(3) If G is a LAG with identity element e, its *Lie algebra* is defined as the tangent space at the identity,

$$\mathrm{Lie}(G) := T_e(G).$$

Question. How to compute tangent spaces and Lie algebras in practice?

(1) First consider a polynomial ring $A = k[T_1, \ldots, T_n]$ or a localization of it, such as

$$k[\mathrm{GL}_n] = k[T_{ij}, \det(T_{ij})^{-1}].$$

A derivation $X : A \to A$ is determined by the values $X(T_1), \ldots, X(T_n)$, because X is then determined on polynomials by the Leibniz rule and on fractions by the quotient rule. This gives an isomorphism

$$\begin{aligned} T(\mathrm{Sp}(A)) = \mathrm{Der}_k(A, A) &\xrightarrow{\sim} \bigoplus_{i=1}^n A\frac{\partial}{\partial T_i}, \\ X &\longmapsto \sum_{i=1}^n X(T_i)\frac{\partial}{\partial T_i}. \end{aligned}$$

In the same way, the tangent space at a point $x \in \mathrm{Sp}(A)$ is

$$T_x(\mathrm{Sp}(A)) = \bigoplus_{i=1}^{n} k \, \frac{\partial}{\partial T_i}\Big|_x, \tag{7.1}$$

where $\frac{\partial}{\partial T_i}\big|_x$ denotes the derivation $f \mapsto \frac{\partial f}{\partial T_i}(x)$.

(2) For a general affine variety, the following recipe applies. Write $B = A/I$ with $A = k[T_1, \dots, T_n]$ and $I = (f_1, \dots, f_m)$, and let $x \in \mathrm{Sp}(B) = V(I)$. Composition with the quotient map $A \twoheadrightarrow B$ defines an injection

$$\begin{aligned} T_x(\mathrm{Sp}(B)) = \mathrm{Der}_k(B, B/\mathfrak{m}_x B) &\longhookrightarrow \mathrm{Der}_k(A, A/\mathfrak{m}_x) = T_x(\mathrm{Sp}(A)), \\ X &\longmapsto X \circ [A \twoheadrightarrow B]. \end{aligned}$$

Its image consists of those $X = \sum_{i=1}^n x_i \frac{\partial}{\partial T_i}\big|_x$ that factor through A/I, which means $X(I) = 0$. By the Leibniz rule and because $f_j(x) = 0$, this is equivalent to $X(f_j) = 0$ for $j = 1, \dots, m$. We conclude that

$$T_x(\mathrm{Sp}(B)) = \ker\Big(\text{Jacobi matrix } \Big(\frac{\partial f_j}{\partial T_i}(x)\Big)_{j,i} : k^n \longrightarrow k^m\Big). \tag{7.2}$$

In general, one has

$$\dim_k T_x(\mathrm{Sp}(B)) \geq \dim_x(\mathrm{Sp}(B)),$$

with equality if and only if $\mathrm{Sp}(B)$ is smooth in x.

Example 7.4. (1) For $G = \mathrm{GL}_n$, we saw in (7.1) that the Lie algebra is

$$\mathfrak{gl}_n := \mathrm{Lie}(\mathrm{GL}_n) = \mathrm{M}_n(k), \qquad \sum_{i,j} x_{ij} \frac{\partial}{\partial T_{ij}}\Big|_{\mathbb{1}_n} \longmapsto (x_{ij})_{i,j},$$

(2) The special linear group $\mathrm{SL}_n \subseteq \mathrm{GL}_n$ is defined by the single equation $f = \det(T_{ij}) - 1$. The gradient of $\det(T_{ij})$ evaluated at $\mathbb{1}_n$ is the linear form $x \mapsto \sum_i x_{ii} = \mathrm{tr}(x)$ on $\mathfrak{gl}_n$, and (7.2) gives

$$\mathfrak{sl}_n := \mathrm{Lie}(\mathrm{SL}_n) = \{x \in \mathfrak{gl}_n \mid \mathrm{tr}(x) = 0\}.$$

(3) Let $H \in \mathrm{M}_n(k)$ be symmetric or skew-symmetric with $\det(H) \in k^\times$, and let $G = \mathrm{O}(H)$ resp. $G = \mathrm{Sp}(H)$ be the corresponding orthogonal resp. symplectic group,

$$G = \{x \in \mathrm{M}_n(k) \mid x^t H x = H\}.$$

The n^2 entries of $x^t H x - H$ are the defining equations of G. Applying $\frac{\partial}{\partial T_{ij}}$ to the matrix-valued function $x \mapsto x^t H x - H$ and evaluating at $\mathbb{1}_n$ gives

$$E_{ij}^t H + H E_{ij}$$

by the Leibniz rule, where E_{ij} denotes the elementary matrix with a single entry 1 in position (i, j). So the Jacobi matrix at $\mathbb{1}_n$ is the linear map $x \mapsto x^t H + Hx$, and we obtain

$$\begin{aligned} \mathrm{Lie}(G) &= \Big\{\sum_{i,j} x_{ij} E_{ij} \in \mathfrak{gl}_n \;\Big|\; \Big(\sum_{i,j} x_{ij} E_{ij}\Big)^t H + H\Big(\sum_{i,j} x_{ij} E_{ij}\Big) = 0\Big\} \\ &= \{x \in \mathfrak{gl}_n \mid x^t H + Hx = 0\}. \end{aligned}$$

For example, if $H = \mathbb{1}_n$, then

$$\mathrm{Lie}(\mathrm{SO}_n) = \mathrm{Lie}(\mathrm{O}_n) = \{x \in \mathfrak{gl}_n \mid x^t = -x\}$$

is the space of skew-symmetric matrices. Here, $\mathrm{Lie}(\mathrm{SO}_n) = \mathrm{Lie}(\mathrm{O}_n)$ because $\mathrm{SO}_n = \mathrm{O}_n^\circ$ is open in O_n. Since LAGs are smooth, we obtain in particular that

$$\dim \mathrm{SO}_n = \frac{n(n-1)}{2}.$$

7.3. **Lie algebras.** Let $G = \mathrm{Sp}(A)$ be a LAG. Our goal is to define a Lie bracket on $\mathrm{Lie}(G)$ to make it into an actual "Lie algebra":

Definition 7.5. A *Lie algebra* over k is a k-vector space V together with an alternating bilinear pairing $[\ ,\ ] : V \times V \to k$, called its *Lie bracket*, such that the *Jacobi identity* holds:

$$[x,[y,z]] + [y,[z,x]] + [z,[x,y]] = 0.$$

We begin with a series of observations.

• A vector field can be evaluated at a point $x \in G$. This defines the map

$$\begin{aligned} T(G) = \mathrm{Der}_k(A,A) &\longrightarrow \mathrm{Der}_k(A, A/\mathfrak{m}_x) = T_x(G), \\ X &\longmapsto [A \twoheadrightarrow A/\mathfrak{m}_x] \circ X. \end{aligned}$$

For $x = e$, the target is $\mathrm{Lie}(G)$.

• The vector fields form a Lie algebra with Lie bracket given by the commutator

$$[\ ,\ ] : T(G) \times T(G) \longrightarrow T(G), \qquad [X,Y](f) := X(Y(f)) - Y(X(f)).$$

This is a general construction: Given any associative, not necessarily unital or commutative k-algebra C, the commutator $[X,Y] := XY - YX$ makes C into a Lie algebra. In our setting, we take $C = \mathrm{End}_k(A)$. One only has to check (exercise) that the commutator of vector fields is again a vector field.

• Vector fields and tangent vectors have functoriality properties. If $\phi : G \xrightarrow{\sim} G$ is an automorphism of varieties with pullback $\phi^* : A \to A$, $f \mapsto f \circ \phi$, then it acts on vector fields by

$$\phi_*(X) := (\phi^*)^{-1} \circ X \circ \phi^*, \qquad \text{i.e.} \qquad \phi_*(X)(f) = X(f \circ \phi) \circ \phi^{-1}.$$

This only makes sense for isomorphisms because we have to be able to invert ϕ^*. Individual tangent vectors, however, can always be pushed forward. A morphism $\phi : \mathrm{Sp}(B) \to \mathrm{Sp}(A)$ with $\phi(y) = x$ induces the *differential*

$$\begin{aligned} d_y\phi : T_y(\mathrm{Sp}(B)) &\longrightarrow T_x(\mathrm{Sp}(A)), \\ X_y &\longmapsto X_y \circ \phi^*. \end{aligned}$$

• Conjugation defines a group homomorphism $G \to \mathrm{Aut}(G)$, $x \mapsto c_x$, where $c_x(y) = xyx^{-1}$ (Example 2.7). Since c_x is an automorphism of varieties with $c_x(e) = e$, the above recipe provides actions of G on $T(G)$ and on $\mathrm{Lie}(G)$.

Definition 7.6. The *adjoint representation* of G is the action of G on $\mathrm{Lie}(G)$ by the differentials of the conjugation maps,

$$\mathrm{Ad} : G \longrightarrow \mathrm{GL}(\mathrm{Lie}(G)), \qquad \mathrm{Ad}(x)(X) := d_e c_x(X) = X \circ c_x^*.$$

In other words, $\mathrm{Ad}(x)(X)(f) = X(f \circ c_x)$ for all $f \in k[G]$. Similarly, G acts on $T(G)$ by $X \mapsto (c_x)_*(X)$, and the evaluation map $T(G) \to \mathrm{Lie}(G)$ is compatible with the two actions.

Lemma 7.7. *Identify* $\mathfrak{gl}_n = \mathrm{M}_n(k)$ *via* $\sum_{i,j} X_{ij} \frac{\partial}{\partial T_{ij}}\big|_{\mathbb{1}_n} \leftrightarrow (X_{ij})_{i,j}$ *as before. Then for* $x \in \mathrm{GL}_n$ *and* $X \in \mathfrak{gl}_n$,

$$\mathrm{Ad}(x)(X) = xXx^{-1}.$$

Consequently, if $G \subseteq \mathrm{GL}_n$ *is a closed subgroup, then* $\mathfrak{g} := \mathrm{Lie}(G) \subseteq \mathfrak{gl}_n$ *is a linear subspace, and* $\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g})$ *is given by conjugation of matrices.*

Proof. Let T_{kl} be a coordinate function. Then

$$(T_{kl} \circ c_x)(y) = (xyx^{-1})_{kl} = \sum_{i,j} x_{ki} y_{ij} (x^{-1})_{jl},$$

so

$$\mathrm{Ad}(x)(X)(T_{kl}) = X(T_{kl} \circ c_x) = \sum_{i,j} x_{ki} X_{ij} (x^{-1})_{jl} = (xXx^{-1})_{kl}.$$

Since a tangent vector at $\mathbb{1}_n$ is determined by its values on the T_{kl}, this proves the formula. For the second statement, note that $T_e(G) \hookrightarrow T_e(\mathrm{GL}_n)$ and that this inclusion is compatible with the differentials of c_x on G and on GL_n. □

Definition 7.8. For vector fields, we have *left translation* as second natural operation. For $x \in G$, let $L_x : G \to G$, $y \mapsto xy$, denote left translation. Taking up our terminology from §3.1, we have

$$\lambda_x := (L_x^*)^{-1} : k[G] \longrightarrow k[G], \qquad (\lambda_x f)(y) = f(x^{-1}y).$$

For a vector field $X \in T(G)$, we define

$$\lambda_x(X) := \lambda_x \circ X \circ \lambda_x^{-1} = (L_x)_*(X).$$

The vector field X is called *left invariant* if $\lambda_x(X) = X$ for all $x \in G$. We denote by

$$\mathscr{L}(G) \subseteq T(G)$$

the subspace of left invariant vector fields.

Since $\lambda_x(X)$ is obtained by conjugating X with the ring automorphism λ_x, we have $\lambda_x([X,Y]) = [\lambda_x(X), \lambda_x(Y)]$. In particular, $\mathscr{L}(G)$ is stable under the Lie bracket.

Example 7.9. Let $G = \mathbb{G}_a$, so $k[G] = k[t]$, and let $X = p(t)\frac{\partial}{\partial t}$ be a vector field. For $x \in k$ and $f \in k[t]$, put $g := \lambda_x^{-1}(f)$, so that $g(t) = f(t+x)$ and $g'(t) = f'(t+x)$. Then

$$\lambda_x(X)(f) = \lambda_x\big(X(g)\big) = \lambda_x\big(p(t)g'(t)\big) = p(t-x)g'(t-x) = p(t-x)f'(t).$$

In other words, $\lambda_x(X) = p(t-x)\frac{\partial}{\partial t}$ is the vector field X shifted by x. Thus X is left invariant if and only if $p(t-x) = p(t)$ for all $x \in k$, which holds if and only if p is constant because k is infinite. So $\mathscr{L}(\mathbb{G}_a) = k \cdot \frac{\partial}{\partial t}$ is one-dimensional, matching the following proposition.

Proposition 7.10. *The evaluation map $T(G) \to T_e(G)$, $X \mapsto X_e$, restricts to an isomorphism*

$$\mathscr{L}(G) \xrightarrow{\sim} \mathrm{Lie}(G).$$

Transporting the Lie bracket of $\mathscr{L}(G)$ along this isomorphism endows $\mathrm{Lie}(G)$ with the structure of a Lie algebra.

Proof. Injectivity. Let $X \in \mathscr{L}(G)$. For $f \in k[G]$ and $x \in G$, we compute

$$X(f)(x) = \big(\lambda_x^{-1}(X(f))\big)(e) \overset{X \in \mathscr{L}(G)}{=} \big(X(\lambda_x^{-1} f)\big)(e) = X_e(\lambda_x^{-1} f).$$

Here, we used $(\lambda_x^{-1}h)(e) = h(xe) = h(x)$ in the first step and $\lambda_x^{-1} \circ X = X \circ \lambda_x^{-1}$ in the second. So X is determined by X_e, which means that a given tangent vector has at most one left invariant extension.

Surjectivity. Let $X_e \in \mathrm{Lie}(G)$ be given, viewed as a derivation $X_e : A \to k$. Consider the comultiplication $m^* : A \to A \otimes_k A$ and define

$$X := (\mathrm{id} \otimes X_e) \circ m^* : A \xrightarrow{m^*} A \otimes_k A \xrightarrow{\mathrm{id} \otimes X_e} A.$$

This is a ring map composed with a derivation, and hence a derivation. In function notation, if $f \in A$ satisfies

$$f(xy) = \sum_{i=1}^{r} p_i(x) q_i(y) \qquad \text{for } p_i, q_i \in k[G],$$

i.e. $m^*(f) = \sum_i p_i \otimes q_i$, then

$$X(f)(x) = \sum_{i=1}^{r} p_i(x) X_e(q_i) = X_e\Big(\sum_{i=1}^{r} p_i(x) q_i\Big) = X_e(\lambda_x^{-1} f).$$

Setting $x = e$ shows that X evaluates to X_e at the identity. Moreover, the formula $X(f)(x) = X_e(\lambda_x^{-1} f)$ shows that X is left invariant: For $z \in G$,

$$\big(\lambda_z \circ X \circ \lambda_z^{-1}\big)(f)(x) = X(\lambda_z^{-1} f)(z^{-1}x) = X_e\big(\lambda_{z^{-1}x}^{-1} \lambda_z^{-1} f\big) = X_e(\lambda_x^{-1} f) = X(f)(x),$$

where we used $\lambda_{z^{-1}x}^{-1} \circ \lambda_z^{-1} = \lambda_{x^{-1}z} \circ \lambda_{z^{-1}} = \lambda_{x^{-1}}$. So $X \in \mathscr{L}(G)$ is a left invariant extension of X_e. □

Summary. We have the following picture:

$$\begin{array}{ccccc} T(G) & \supseteq & \mathscr{L}(G) & [\ ,\ ] & \text{defined directly,} \\ \big\downarrow & & \big\downarrow \cong & \big\downarrow & \\ T_e(G) & = & \mathrm{Lie}(G) & [\ ,\ ] & \text{transported.} \end{array}$$

In this way, $\mathrm{Lie}(G)$ becomes a Lie algebra: Its bracket $[\ ,\ ]$ is bilinear, alternating, and satisfies the Jacobi identity.

References

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